

# OCS Commands

Complete desk reference covering 443 command tokens across 5 distinct sections: Core OCS 2D Drafting Commands & Aliases (206), OCS System Variables (96), Duplicate Registered Command Aliases (74), Application & Clipboard Meta-Commands (39), and Recognized 3D/Mesh Sub-Commands (28).

**Legend:**  1-Key Priority Shortcut  Standard ocad.pgp Alias  Full Command (No Default Shortcut)

# ≡ Alphabetical Command Index



Select any of the 443 documented tokens below to jump directly to its reference entry.

|                 |                     |                     |                      |                |                       |                    |                       |
|-----------------|---------------------|---------------------|----------------------|----------------|-----------------------|--------------------|-----------------------|
| #               |                     | B                   |                      | D              |                       | E                  |                       |
| 3DARRAY         | 3D / mesh           | B                   | Alias for BLOCK      | D              | Alias for DIMSTYLE    | E                  | Alias for ERASE       |
| 3DFACE          | 3D / mesh           | BATTMAN             | Blocks & References  | DAL            | Alias for DIMALIGNED  | EATTEXT            | Blocks & References   |
| 3DMESH          | 3D / mesh           | BATTMAN             | Blocks & References  | DAN            | Alias for DIMANGULAR  | ED                 | Alias for TEXTEDIT    |
| 3DMOVE          | 3D / mesh           | BCOUNT              | Blocks & References  | DATAEXTRACTION | Inquiry & Measure     | EDGESURF           | 3D / mesh             |
| 3DORBIT         | View & Navigation   | BE                  | Alias for BEDIT      | DATAEXTRACTION | Annotation            | EL                 | Alias for ELLIPSE     |
| 3DPOLY          | Draw                | BEDIT               | Blocks & References  | DATALINK       | Annotation            | ELEVATION          | System variable       |
| 3DPOLY          | 3D / mesh           | BLE                 | Alias for BLEND      | DBA            | Alias for DIMBASELINE | ELLIPSE            | Draw                  |
| 3DROTATE        | 3D / mesh           | BLEND               | Modify               | DCE            | Alias for DIMCENTER   | ERASE              | Modify                |
| 3DSCALE         | 3D / mesh           | BLOCK               | Blocks & References  | DCO            | Alias for DIMCONTINUE | ETRANSMIT          | System & Utilities    |
| 30              | Alias for 3DORBIT   | BOUNDARY            | Draw                 | DDI            | Alias for DIMDIAMETER | EX                 | Alias for EXTEND      |
| A               |                     | BOUNDARY            | Draw                 | DDIM           | Alias for DIMSTYLE    | EXIT               | Application command   |
| A               | Alias for ARC       | BOX                 | 3D Modeling          | DE             | Alias for DDEDIT      | EXPLODE            | Modify                |
| ABOUT           | Application command | BR                  | Alias for BREAK      | DELOBJ         | System variable       | EXTEND             | Modify                |
| ADDSELECTED     | Application command | BREAK               | Modify               | DESELALL       | Application command   | EXTRIM             | Modify                |
| AI_EDITCUSTFILE | System & Utiliti... | BURST               | Modify               | DESELECT       | Application command   | EXTRUDE            | 3D Modeling           |
| ALIASEDIT       | System & Utilities  | C                   |                      | DGNFRAME       | System variable       | F                  |                       |
| ALIGN           | Modify              | C                   | Alias for CIRCLE     | DI             | Alias for DIST        | F                  | Alias for FILLET      |
| ANGBASE         | System variable     | CASCADE             | Application command  | DIMALIGNED     | Dimension             | FACETRES           | System variable       |
| ANGDIR          | System variable     | CDATE               | System variable      | DIMANGULAR     | Dimension             | FIELD              | Annotation            |
| APERTURE        | System variable     | CDATE               | System variable      | DIMARC         | Dimension             | FILEDIA            | System variable       |
| APpload         | System & Utilities  | CELTSCALE           | System variable      | DIMASSOC       | System variable       | FILLET             | Modify                |
| APpload         | System & Utilities  | CENTERLINE          | Dimension            | DIMBASELINE    | Dimension             | FILTER             | Modify                |
| AR              | Alias for ARRAY     | CENTERMARK          | Dimension            | DIMBREAK       | Dimension             | FIND               | Annotation            |
| ARC             | Draw                | CH                  | Alias for PROPERTIES | DIMCLRD        | System variable       | FIND               | Annotation            |
| AREA            | Inquiry & Measure   | CHA                 | Alias for CHAMFER    | DIMCLRE        | System variable       | FLATSHOT           | 3D Modeling           |
| AREA            | System variable     | CHAMFER             | Modify               | DIMCLRT        | System variable       | FLATSHOT           | 3D Modeling           |
| ARRANGE         | Application command | CHANGELOG           | Application command  | DIMCONTINUE    | Dimension             | FLATSHOT           | 3D / mesh             |
| ARRAY           | Modify              | CHSPACE             | Modify               | DIMDIAMETER    | Dimension             | FLATTEN            | Modify                |
| ARRAY3D         | 3D / mesh           | CIRCLE              | Draw                 | DIMEDIT        | Dimension             | FRAME              | System variable       |
| ARRAYEDIT       | Modify              | CLEAR               | Application command  | DIMLFAC        | System variable       | G                  |                       |
| ATTDEF          | Blocks & References | CLOSE               | Application command  | DIMLINEAR      | Dimension             | GEOGRAPHICLOCATION | System & ...          |
| ATTDEF          | Blocks & References | CLOSEALL            | System & Utilities   | DIMORDINATE    | Dimension             | GRADIENT           | Draw                  |
| ATTSYNC         | Blocks & References | CLOSEALL            | Application command  | DIMOVERRIDE    | Dimension             | GRIDDISPLAY        | System variable       |
| ATTSYNC         | Blocks & References | CLOSEALLOTHER       | System & Utilities   | DIMRADIUS      | Dimension             | GRIDMODE           | System variable       |
| AUDIT           | System & Utilities  | CLOSEALLOTHER       | Application comm...  | DIMREASSOCIATE | Dimension             | GRIDUNIT           | System variable       |
| AUDIT           | System & Utilities  | CLR                 | Application command  | DIMSCALE       | System variable       | GRIPCOLOR          | System variable       |
| AUNITS          | System variable     | CMDDIA              | System variable      | DIMSPACE       | Dimension             | GRIPHOT            | System variable       |
| AUPREC          | System variable     | CMDECHO             | System variable      | DIMSTYLE       | Dimension             | GRIPHOVER          | System variable       |
| AUTOSNAP        | System variable     | CO                  | Alias for COPY       | DIMTXT         | System variable       | GRIPS              | System variable       |
|                 |                     | COMMANDLINEFADETIME | System v...          | DISPSILH       | System variable       | GRIPSIZE           | System variable       |
|                 |                     | COMPARE             | System & Utilities   | DIST           | Inquiry & Measure     | H                  |                       |
|                 |                     | CONVTOMESH          | 3D / mesh            | DIVIDE         | Draw                  | H                  | Alias for HATCH       |
|                 |                     | CONVTOSOLID         | 3D / mesh            | DLI            | Alias for DIMLINEAR   | HATCH              | Draw                  |
|                 |                     | CONVTOSURFACE       | 3D / mesh            | DONUT          | Draw                  | HATCHEDIT          | Modify                |
|                 |                     | COPY                | Modify               | DRA            | Alias for DIMRADIUS   | HATCHTOBACK        | Modify                |
|                 |                     | COPYCLIP            | Application command  | DRAWORDER      | Modify                | HB                 | Alias for HATCHTOBACK |
|                 |                     | COUNT               | Blocks & References  | DSPACE         | Alias for DIMSPACE    | HELIX              | Draw                  |
|                 |                     | CP                  | Alias for COPY       | DT             | Alias for TEXT        | HELP               | Application command   |
|                 |                     | CROSSINGAREACOLOR   | System varia...      | DWFFRAME       | System variable       | HIGHLIGHT          | System variable       |
|                 |                     | CUI                 | System & Utilities   | DWGNAME        | System variable       |                    |                       |
|                 |                     | CUI                 | System & Utilities   | DWGPREFIX      | System variable       |                    |                       |
|                 |                     | CURSORSIZE          | System variable      | DWGPROPS       | System & Utilities    |                    |                       |
|                 |                     | CUTCLIP             | Application command  |                |                       |                    |                       |
|                 |                     | CYLINDER            | 3D Modeling          |                |                       |                    |                       |

## I

|                          |                     |
|--------------------------|---------------------|
| <b>I</b>                 | Alias for INSERT    |
| <b>ID</b>                | Inquiry & Measure   |
| <b>ID</b>                | Alias for ID        |
| <b>IMAGEFRAME</b>        | System variable     |
| <b>INSERT</b>            | Blocks & References |
| <b>INSUNITS</b>          | System variable     |
| <b>INSUNITSDEFSOURCE</b> | System varia...     |
| <b>INSUNITSDEFTARGET</b> | System varia...     |
| <b>INTERFERE</b>         | 3D / mesh           |
| <b>INTERSECT</b>         | 3D Modeling         |
| <b>INTERSECT</b>         | 3D / mesh           |
| <b>ISOLINES</b>          | System variable     |

## J

|             |                |
|-------------|----------------|
| <b>J</b>    | Alias for JOIN |
| <b>JOIN</b> | Modify         |

## L

|                    |                     |
|--------------------|---------------------|
| <b>L</b>           | Alias for LINE      |
| <b>LA</b>          | Alias for LAYER     |
| <b>LAYCUR</b>      | Layers & Properties |
| <b>LAYDEL</b>      | Layers & Properties |
| <b>LAYDEL</b>      | Layers & Properties |
| <b>LAYER</b>       | Layers & Properties |
| <b>LAYERS</b>      | Alias for LAYER     |
| <b>LAYFRZ</b>      | Layers & Properties |
| <b>LAYISO</b>      | Layers & Properties |
| <b>LAYMCH</b>      | Layers & Properties |
| <b>LAYMRG</b>      | Layers & Properties |
| <b>LAYMRG</b>      | Layers & Properties |
| <b>LAYOFF</b>      | Layers & Properties |
| <b>LAYON</b>       | Layers & Properties |
| <b>LAYOUTPANEL</b> | Application command |
| <b>LAYOUTTAB</b>   | Application command |
| <b>LAYTHW</b>      | Layers & Properties |
| <b>LAYTRANS</b>    | Layers & Properties |
| <b>LAYUNISO</b>    | Layers & Properties |
| <b>LAYWALK</b>     | Layers & Properties |
| <b>LE</b>          | Alias for QLEADER   |
| <b>LENGTHEN</b>    | Modify              |
| <b>LINE</b>        | Draw                |
| <b>LIST</b>        | Inquiry & Measure   |
| <b>LOFT</b>        | 3D Modeling         |
| <b>LOGINNAME</b>   | System variable     |
| <b>LTSCALE</b>     | Layers & Properties |
| <b>LTSCALE</b>     | System variable     |
| <b>LUNITS</b>      | System variable     |
| <b>LUPREC</b>      | System variable     |
| <b>LWDEFAULT</b>   | System variable     |
| <b>LWDISPLAY</b>   | System variable     |

## M

|                    |                       |
|--------------------|-----------------------|
| <b>M</b>           | Alias for MOVE        |
| <b>MASSPROP</b>    | Inquiry & Measure     |
| <b>MASSPROP</b>    | Inquiry & Measure     |
| <b>MATCHPROP</b>   | Layers & Properties   |
| <b>MAXACTVP</b>    | System variable       |
| <b>MEA</b>         | Alias for MEASUREGEOM |
| <b>MEASURE</b>     | Draw                  |
| <b>MEASUREGEOM</b> | Inquiry & Measure     |
| <b>MI</b>          | Alias for MIRROR      |
| <b>MIRROR</b>      | Modify                |
| <b>MIRRTEXT</b>    | System variable       |
| <b>MLD</b>         | Alias for MLEADER     |
| <b>MLEADER</b>     | Annotation            |
| <b>MLEADEREDIT</b> | Annotation            |
| <b>MO</b>          | Alias for PROPERTIES  |
| <b>MOVE</b>        | Modify                |
| <b>MSLTSCALE</b>   | System variable       |
| <b>MT</b>          | Alias for MTEXT       |
| <b>MTEXT</b>       | Annotation            |
| <b>MULTILINE</b>   | Draw                  |
| <b>MULTIPLE</b>    | System & Utilities    |
| <b>MVIEW</b>       | View & Navigation     |

## N

|              |                     |
|--------------|---------------------|
| <b>NCOPY</b> | Modify              |
| <b>NCOPY</b> | Blocks & References |
| <b>NEW</b>   | Application command |

## O

|                  |                     |
|------------------|---------------------|
| <b>O</b>         | Alias for OFFSET    |
| <b>OFFSET</b>    | Modify              |
| <b>OOPS</b>      | Modify              |
| <b>OPEN</b>      | Application command |
| <b>OPTIONS</b>   | System & Utilities  |
| <b>ORTHOMODE</b> | System variable     |
| <b>OSMODE</b>    | System variable     |
| <b>OVERKILL</b>  | Modify              |

## P

|                      |                      |
|----------------------|----------------------|
| <b>P</b>             | Alias for PAN        |
| <b>PAGESETUP</b>     | System & Utilities   |
| <b>PAN</b>           | View & Navigation    |
| <b>PASTE</b>         | Application command  |
| <b>PASTEBLOCK</b>    | Application command  |
| <b>PASTECLIP</b>     | Application command  |
| <b>PASTEORIG</b>     | Application command  |
| <b>PDFFRAME</b>      | System variable      |
| <b>PDFIMPORT</b>     | Blocks & References  |
| <b>PEDIT</b>         | Modify               |
| <b>PERF</b>          | Application command  |
| <b>PERIMETER</b>     | System variable      |
| <b>PICKADD</b>       | System variable      |
| <b>PICKAUTO</b>      | System variable      |
| <b>PICKBOX</b>       | System variable      |
| <b>PICKDRAG</b>      | System variable      |
| <b>PICKFIRST</b>     | System variable      |
| <b>PL</b>            | Alias for PLINE      |
| <b>PLAN</b>          | View & Navigation    |
| <b>PLINE</b>         | Draw                 |
| <b>PLINEGEN</b>      | System variable      |
| <b>PLUGINMANAGER</b> | Application comm...  |
| <b>PLUGINS</b>       | Application command  |
| <b>PO</b>            | Alias for POINT      |
| <b>POINT</b>         | Draw                 |
| <b>POL</b>           | Alias for POLYGON    |
| <b>POLARANG</b>      | System variable      |
| <b>POLARMODE</b>     | System variable      |
| <b>POLYGON</b>       | Draw                 |
| <b>PR</b>            | Alias for PROPERTIES |
| <b>PRESSPULL</b>     | 3D Modeling          |
| <b>PRESSPULL</b>     | 3D / mesh            |
| <b>PRINTALL</b>      | Application command  |
| <b>PRODUCT</b>       | System variable      |
| <b>PROPERTIES</b>    | Layers & Properties  |
| <b>PROPS</b>         | Alias for PROPERTIES |
| <b>PSLTSCALE</b>     | System variable      |
| <b>PU</b>            | Alias for PURGE      |
| <b>PUBLISH</b>       | System & Utilities   |
| <b>PURGE</b>         | System & Utilities   |

## Q

|                   |                     |
|-------------------|---------------------|
| <b>QC</b>         | Alias for QUICKCALC |
| <b>QDIM</b>       | Dimension           |
| <b>QS</b>         | Alias for QSAVE     |
| <b>QSAVE</b>      | Application command |
| <b>QSELECT</b>    | Modify              |
| <b>QUICKCALC</b>  | System & Utilities  |
| <b>QUICKPRINT</b> | Application command |
| <b>QUIT</b>       | Application command |

## R

|                   |                    |
|-------------------|--------------------|
| <b>RAY</b>        | Draw               |
| <b>RAY</b>        | Alias for RAY      |
| <b>REC</b>        | Alias for RECTANG  |
| <b>RECOVER</b>    | System & Utilities |
| <b>RECOVERALL</b> | System & Utilities |
| <b>RECTANG</b>    | Draw               |
| <b>REGEN</b>      | View & Navigation  |
| <b>REGENALL</b>   | View & Navigation  |
| <b>REGENMODE</b>  | System variable    |
| <b>REGION</b>     | Draw               |
| <b>REINIT</b>     | System & Utilities |
| <b>REINIT</b>     | System & Utilities |
| <b>REN</b>        | Alias for RENAME   |
| <b>REVCLLOUD</b>  | Draw               |
| <b>REVOLVE</b>    | 3D Modeling        |
| <b>REVOLVE</b>    | 3D / mesh          |
| <b>REVSURF</b>    | 3D / mesh          |
| <b>RO</b>         | Alias for ROTATE   |
| <b>ROTATE</b>     | Modify             |
| <b>ROTATE3D</b>   | 3D / mesh          |
| <b>RULESURF</b>   | 3D / mesh          |

| S                           |                     | T                   |                     | W                      |                     |
|-----------------------------|---------------------|---------------------|---------------------|------------------------|---------------------|
| <b>S</b>                    | Alias for STRETCH   | <b>T</b>            | Alias for MTEXT     | <b>WB</b>              | Alias for WBLOCK    |
| <b>SA</b>                   | Alias for SAVEAS    | <b>TABLE</b>        | Annotation          | <b>WBLOCK</b>          | Blocks & References |
| <b>SAVE</b>                 | Application command | <b>TABLEEXPORT</b>  | Annotation          | <b>WINDOWAREACOLOR</b> | System variable     |
| <b>SAVEALL</b>              | System & Utilities  | <b>TABSURF</b>      | 3D / mesh           | <b>WIPEOUT</b>         | Draw                |
| <b>SAVEALL</b>              | Application command | <b>TCOUNT</b>       | Annotation          | <b>WIPEOUTFRAME</b>    | System variable     |
| <b>SAVEAS</b>               | Application command | <b>TCOUNT</b>       | Annotation          |                        |                     |
| <b>SAVETIME</b>             | System variable     | <b>TEXT</b>         | Annotation          |                        |                     |
| <b>SC</b>                   | Alias for SCALE     | <b>TEXTFILL</b>     | System variable     | <b>X</b>               | Alias for EXPLODE   |
| <b>SCALE</b>                | Modify              | <b>TEXTMASK</b>     | Annotation          | <b>XA</b>              | Alias for XATTACH   |
| <b>SCRIPT</b>               | System & Utilities  | <b>TEXTQLTY</b>     | System variable     | <b>XATTACH</b>         | Blocks & References |
| <b>SECTION</b>              | 3D Modeling         | <b>TEXTTTOFRONT</b> | Modify              | <b>XB</b>              | Alias for XBIND     |
| <b>SECTION</b>              | 3D / mesh           | <b>TEXTTTOFRONT</b> | Annotation          | <b>XCLIP</b>           | Blocks & References |
| <b>SECTIONPLANE</b>         | 3D Modeling         | <b>THICKNESS</b>    | System variable     | <b>XCLIPFRAME</b>      | System variable     |
| <b>SELECTALL</b>            | Application command | <b>TILEHORZ</b>     | Application command | <b>XL</b>              | Alias for XLINE     |
| <b>SELECTIONAREA</b>        | System variable     | <b>TILEVERT</b>     | Application command | <b>XLINE</b>           | Draw                |
| <b>SELECTIONAREAOPACITY</b> | System ...          | <b>TIME</b>         | Inquiry & Measure   | <b>XOPEN</b>           | Blocks & References |
| <b>SELECTIONEFFECT</b>      | System variable     | <b>TR</b>           | Alias for TRIM      | <b>XR</b>              | Alias for XREF      |
| <b>SELECTIONEFFECTCOLOR</b> | System ...          | <b>TRACE</b>        | 3D / mesh           | <b>XREF</b>            | Blocks & References |
| <b>SELECTIONPREVIEW</b>     | System variable     | <b>TRIM</b>         | Modify              |                        |                     |
| <b>SELECTSIMILAR</b>        | Modify              | <b>TS</b>           | Alias for STYLE     |                        |                     |
| <b>SETBYLAYER</b>           | Layers & Properties | <b>TXT2MTXT</b>     | Annotation          |                        |                     |
| <b>SETVAR</b>               | System & Utilities  |                     |                     | <b>ZOOM</b>            | View & Navigation   |
| <b>SHEETSET</b>             | System & Utilities  |                     |                     | <b>ZOOMFACTOR</b>      | System variable     |
| <b>SHORTCUTS</b>            | Application command |                     |                     | <b>ZOOMWHEEL</b>       | System variable     |
| <b>SKETCH</b>               | Draw                |                     |                     | <b>ZS</b>              | Alias for ZOOM      |
| <b>SNAPANG</b>              | System variable     |                     |                     |                        |                     |
| <b>SNAPISOPAIR</b>          | System variable     |                     |                     |                        |                     |
| <b>SNAPMODE</b>             | System variable     |                     |                     |                        |                     |
| <b>SNAPTYPE</b>             | System variable     |                     |                     |                        |                     |
| <b>SNAPUNIT</b>             | System variable     |                     |                     |                        |                     |
| <b>SO</b>                   | 3D / mesh           |                     |                     |                        |                     |
| <b>SOLID2D</b>              | 3D / mesh           |                     |                     |                        |                     |
| <b>SPELL</b>                | Annotation          |                     |                     |                        |                     |
| <b>SPL</b>                  | Alias for SPLINE    |                     |                     |                        |                     |
| <b>SPLINE</b>               | Draw                |                     |                     |                        |                     |
| <b>STARTUP</b>              | System variable     |                     |                     |                        |                     |
| <b>STEPEXPORT</b>           | 3D Modeling         |                     |                     |                        |                     |
| <b>STLOUT</b>               | 3D Modeling         |                     |                     |                        |                     |
| <b>STRETCH</b>              | Modify              |                     |                     |                        |                     |
| <b>STYLE</b>                | Annotation          |                     |                     |                        |                     |
| <b>SUBTRACT</b>             | 3D Modeling         |                     |                     |                        |                     |
| <b>SUBTRACT</b>             | 3D / mesh           |                     |                     |                        |                     |
| <b>SURFTYPE</b>             | System variable     |                     |                     |                        |                     |
| <b>SURFU</b>                | System variable     |                     |                     |                        |                     |
| <b>SURFV</b>                | System variable     |                     |                     |                        |                     |
| <b>SWEEP</b>                | 3D Modeling         |                     |                     |                        |                     |
| <b>SWEEP</b>                | 3D / mesh           |                     |                     |                        |                     |
| <b>SYSVARMONITOR</b>        | System & Utilities  |                     |                     |                        |                     |

## 1K The 1-Key Priority Aliases (A-Z Direct Keys)

19 Single-Key Commands

|                   |                 |                 |
|-------------------|-----------------|-----------------|
| <b>A</b> ARC      | <b>B</b> BLOCK  | <b>C</b> CIRCLE |
| <b>D</b> DIMSTYLE | <b>E</b> ERASE  | <b>F</b> FILLET |
| <b>H</b> HATCH    | <b>I</b> INSERT | <b>J</b> JOIN   |

**L** LINE

**M** MOVE

**O** OFFSET

**P** PAN

**S** STRETCH

**V** VIEW

**V** VIEW

**W** WBLOCK

**X** EXPLODE

**Z** ZOOM

## **Fn** Drafting Toggles & Function Keys (F1 – F12)

**F1**

Help

Displays contextual Help documentation for active command or window.

**F2**

Text Window

Expands the Command Line history log window.

**F3**

OSNAP Toggle

Toggles Running Object Snap (Endpoint, Midpoint, Center, Intersection).

**F4**

3D OSNAP

Toggles 3D Object Snap modes for solid faces, edges, and vertices.

**F5**

Isoplane

Cycles between 2D Isometric planes (Top, Left, Right).

**F6**

Dynamic UCS

Toggles automatic alignment of the UCS to planar solid faces.

**F7**

Grid Display

Toggles background drafting grid on or off.

**F8**

Ortho Mode

Constrains cursor movement strictly to 90° horizontal and vertical axes.

**F9**

Grid Snap

Constrains cursor movement to discrete grid intervals.

**F10**

Polar Tracking

Guides cursor along specified incremental angle vectors.

**F11**

Object Snap Tracking

Tracks cursor along alignment paths from existing object snap points.

**F12**

Dynamic Input

Displays command prompts, coordinate inputs, and dimensions near the cursor.

## **^** Control Accelerators & System Toggles

**Ctrl + 0**

Clean Screen

Hides toolbars, ribbons, and palettes to maximize drawing area canvas.

**Ctrl + 1**

Properties Palette

Opens or closes the object Properties inspector palette.

**Ctrl + 2**

DesignCenter

Opens OCS DesignCenter (browse blocks, styles, layers across files).

**Ctrl + 3**

Tool Palettes

Opens the Tool Palettes window with categorized drafting symbols.

**Ctrl + 4****Sheet Set Manager**

Opens the Sheet Set Manager palette to organize multi-sheet drawings.

**Ctrl + 8****QuickCalc**

Opens the QuickCalc interactive calculation palette.

**Ctrl + 9****Command Line**

Toggles the Command Line prompt bar visibility on or off.

**Ctrl + Shift + C****Copy with Base Point**

Copies selected objects to clipboard with an explicit user base point.

**Ctrl + Shift + V****Paste as Block**

Pastes copied clipboard objects directly as a new anonymous block.

**Ctrl + S****Save**

Saves current drawing (QSAVE).

**Ctrl + P****Plot / Print**

Opens the Plot dialog to publish or print sheets to PDF or plotter.

**Ctrl + Z****Undo**

Reverses the most recent action or command.

**Ctrl + Y****Redo**

Re-applies the most recent undone action.

**Ctrl + A****Select All**

Selects all unlocked objects in current model or paper space.

## Complete Categorized OCS Commands Reference

Organized by drafting discipline. Shows keyboard shortcuts, full OCS command names, ocad.pgp syntax, and practical tips.

**L** LINE

Creates a series of contiguous straight 2D or 3D line segments.

*Press Spacebar or Enter to repeat. Right-click or Enter to exit. Type U while active to undo last vertex.*

**PL** PLINE

Creates a 2D polyline, a single compound object composed of joined line and arc segments with uniform or tapering width.

*Switch between arc and line sub-modes while drawing by typing A or L. Use Close (C) to form watertight geometry.*

**C** CIRCLE

Creates a circle using center-radius, center-diameter, 2-point, 3-point, or Tangent-Tangent-Radius (TTR).

*Default prompt asks for center and radius; enter D to specify diameter instead. Use 2P for fitting between two parallel lines.*

**A** ARC

Creates an arc using 11 different geometric methods (3-point, Start-Center-End, Start-End-Radius, etc.).

*By default, OCS draws arcs counter-clockwise. Hold Ctrl while dragging to invert the arc direction.*

**REC** RECTANG

Creates a rectangular closed polyline from two opposing corner points.

*Type D after placing the first corner to input exact Length and Width dimensions before clicking the quadrant.*

**POL** POLYGON

Creates an equilateral closed polyline with 3 to 1024 sides.

*Specify Inscribed in circle (I) to measure radius to vertices, or Circumscribed (C) to measure radius to edge midpoints.*

**EL** ELLIPSE

Creates an ellipse or an elliptical arc using major and minor axes.

*Type A to create an elliptical arc. Set PELLIPSE system variable to 1 if you want the ellipse created as a true polyline.*

**H** HATCH

Fills an enclosed area or selected objects with a hatch pattern, solid color fill, or gradient.

*Always ensure your boundary is cleanly closed. Use "Pick Points" for internal islands or "Select" to use existing closed polylines.*

**GD** GRADIENT

Fills an enclosed boundary with a two-color or one-color smooth gradient shading.

*Used extensively for architectural diagrams, solar shading, and presentation graphics.*

**SPL** SPLINE

Creates a smooth curve that passes through or near a sequence of control points (NURBS curve).

*Toggle Method between "Fit" (curve passes directly through points) and "Control Vertices" (CVs pull the curve smoothly).*

#### **XL** **XLINE**

Creates an infinite construction line in both directions.

*Sub-options include H (Horizontal), V (Vertical), A (Angle), B (Bisect angle), and O (Offset distance).*

#### **Rec: RY** **RAY**

Creates a semi-infinite construction line extending infinitely in one direction from a starting point.

*Great for sightlines, radial sun-path diagrams, and projection radiating from a single station point.*

#### **ML** **MULTILINE**

Creates multiple parallel lines simultaneously according to an active multiline style.

*Set multiline styles via MLSTYLE. Adjust Justification (Top, Zero, Bottom) and Scale to match wall thickness.*

#### **P0** **POINT**

Creates a single point marker object at specified XYZ coordinates.

*Run DDPTYPE to change point visual style from invisible single dot to crosshairs, circles, or X marks.*

#### **D0** **DONUT**

Draws a filled circle or solid wide ring by specifying inside and outside diameters.

*Set inside diameter to 0 to create a solid filled circular dot (commonly used for electrical terminal points).*

#### **Rec: RC** **REVLOUD**

Creates a revision cloud composed of contiguous arc segments to highlight drawing changes.

*Use the "Arc length" option to scale the cloud scallops to match the sheet viewport scale before placing.*

#### **Rec: HLX** **HELIX**

Creates a 2D spiral or 3D spring/helix curve.

*Specify base radius, top radius, number of turns, and height. Used as paths for SWEEP to create threads and springs.*

#### **Rec: SK** **SKETCH**

Creates a series of freehand line segments as you drag the cursor.

*Set SKPOLY system variable to 1 so the resulting freehand paths are polylines rather than thousands of individual lines.*

#### **Rec: W0** **WIPEOUT**

Creates a polygonal background mask object that conceals underlying objects with the current background color.

*Run WIPEOUT -> Frames -> Off before final plotting to hide boundary outlines while keeping objects concealed underneath.*

#### **B0** **BOUNDARY**

Creates a closed polyline or region from an enclosed area by clicking an internal point.

*The cleanest way to generate a perimeter polyline from crossing, overlapping line geometry without manual tracing.*

#### **REG** REGION

Converts closed 2D shapes (lines, polylines, circles) into a 2D surface region object.

*Regions support boolean operations (UNION, SUBTRACT, INTERSECT) and calculation of mass/area properties.*

#### **DIV** DIVIDE

Places evenly spaced point objects or blocks along the length of an entity.

*Does not physically cut the object; it inserts point nodes. Use the Block (B) option to insert repeating symbols along a curve.*

#### **ME** MEASURE

Places point objects or blocks at specified distance intervals along an entity.

*Unlike DIVIDE (which divides by count), MEASURE divides by length of segment. Selection end determines starting point.*

#### **3P** 3DPOLY

Creates a 3D polyline with vertices that can have independent Z elevation coordinates.

*Unlike standard 2D PLINE, 3D polylines only support straight segments (no arcs) but can freely traverse 3D space.*

#### **B0** BOUNDARY

Creates a region or closed polyline automatically from an enclosed area by clicking an internal point.

*Far faster than tracing complex overlapping geometry. Similar to the pick-point behavior in HATCH.*

**M MOVE**

Moves selected objects a specified distance in a specified direction.

*Use object snaps (OSNAP) for precision basepoints. Type @10,0 to move exactly 10 units right along the X axis.*

**CO COPY**

Duplicates selected objects across one or multiple destination points (also alias CP).

*OCS defaults to multiple copy mode. Type A while copying to quickly create an array of copies along a line.*

**RO ROTATE**

Rotates objects around a base point by angle or reference angle.

*Use the Reference (R) sub-option to align an object to an existing angled geometry without calculating the angle manually.*

**SC SCALE**

Enlarges or reduces selected objects proportionally in X, Y, and Z axes.

*Use Reference (R) option: pick an existing distance and type the desired new target length. Type C to scale a copy.*

**S STRETCH**

Stretches lines, polylines, dimensions, and curves crossing a crossing window.

*CRITICAL: You must use a Green Crossing Window (drag right-to-left) across the vertices you wish to move.*

**TR TRIM**

Trims objects to meet the cutting edges of other objects.

*In modern OCS Quick Mode, simply click or fence-drag across any segment. Hold Shift while trimming to switch to Extend.*

**EX EXTEND**

Extends objects to meet the boundary edges of other objects.

*Hold Shift while extending to instantly switch into Trim mode without exiting the command.*

**F FILLET**

Rounds or fillets the edges of two intersecting objects with a specified radius.

*Type R -> 0 for a sharp 90-degree corner cleanup between crossing or disconnected lines. Use P to fillet an entire polyline at...*

**CHA CHAMFER**

Bevels or chamfers the corners of two intersecting lines by distances or angle.

*Specify distance with D. Distance 0,0 produces an instant sharp corner just like Fillet R 0.*

**O OFFSET**

Creates concentric circles, parallel lines, and parallel curves at a designated offset distance.

Use "Through" (T) option to offset through a specific cursor point rather than typing a numeric distance. Use E to erase source.

## MI MIRROR

Creates a mirrored duplicate of selected objects across a 2-point reflection axis.

*Set MIRRTEXT system variable to 0 to prevent text from being mirrored backwards during reflection.*

## X EXPLODE

Breaks a compound object (block, polyline, hatch, dimension, table) into its component elements.

*Warning: Exploding attributed blocks turns attributes into unassigned raw tag definitions; use BURST instead to preserve...*

## J JOIN

Combines collinear lines, arcs, polylines, and splines into a single unified object.

*Objects must have touching endpoints or be collinear. Quickly turns fragmented lines into a single continuous polyline.*

## BR BREAK

Breaks the selected object between two specified points or splits it at a single point.

*To break an object cleanly at one point, select the object, type F (First point), click the split point, and enter @.*

## AR ARRAY

Creates copies of objects arranged in a pattern: Rectangular, Polar (circular), or along a Path curve.

*Modern arrays are associative: you can click any element later to edit rows, columns, levels, and spacing dynamically.*

## Rec: ARE ARRAYEDIT

Edits associative array objects and their source elements without exploding.

*Allows editing individual items, replacing instances, or adding/deleting rows and columns in existing arrays.*

## AL ALIGN

Aligns objects with other objects in 2D or 3D using 1, 2, or 3 source/destination reference point pairs.

*Combines Move, Rotate, and Scale into a single operation. At the prompt "Scale objects based on alignment points?", select...*

## LEN LENGTHEN

Changes the length of objects and the included angle of arcs.

*Options include Delta (DE), Percent (P), Total (T), and Dynamic (DY). Total lets you set the final exact length directly.*

## PE PEDIT

Edits 2D polylines, joined 3D polylines, and 3D polygon meshes.

*Type M to select multiple lines and convert them all into polylines simultaneously, then type J (Join) with a fuzz distance.*

## HE HATCHEDIT

Modifies an existing hatch pattern, angle, scale, origin point, or boundary associativity.

Double-clicking any hatch pattern in the drawing automatically launches this command or the Hatch Editor contextual ribbon.

## **E** ERASE

Removes selected objects from a drawing.

Type *L* after starting ERASE to delete the Last created object. Type *P* to select Previous selection set.

## **Rec: OP** OOPS

Restores objects erased by the most recent ERASE command without undoing other subsequent drawing commands.

Unlike UNDO (which rolls back everything in sequence), OOPS specifically restores only deleted entities while preserving new...

## **Rec: OK** OVERKILL

Removes duplicate or overlapping lines, arcs, and polylines, and combines partially overlapping or collinear segments.

Indispensable CAD cleanup tool. Dramatically shrinks drawing file size and prevents double-burns in CNC/laser cutting.

## **Rec: FLAT** FLATTEN

Converts 3D geometry and rogue non-zero Z elevations into 2D flat projections on the current UCS plane.

Essential when working with survey or Revit DWG exports where lines sit at varying elevations and refuse to fillet or hatch.

## **Rec: CHS** CHSPACE

Transfers selected objects between Model Space and Paper Space while automatically scaling them to match the viewport.

Allows moving title blocks, notes, or detail callouts between paper space and model space with zero scale distortion.

## **Rec: BST** BURST

Explodes a block while preserving the current visible values of its attributes as standard single/multiline text.

Regular EXPLODE turns attribute values back into tag placeholders (e.g. "REV\_NO"); BURST preserves the actual entered text...

## **Rec: NC** NCOPY

Copies nested objects contained within an Xref or block directly into the current drawing without opening or binding.

Allows you to grab individual boundary lines or equipment geometry from an external background reference instantly.

## **Rec: SS** SELECTSIMILAR

Finds and selects all objects in the drawing that match the properties (layer, color, type, block name) of chosen object.

Type *SE* (Settings) to configure criteria: layer, linetype, block name, color, style, or lineweight.

## **DR** DRAWORDER

Changes the visual display and plotting order of overlapping objects (Bring to Front, Send to Back, Above, Under).

Use "Bring to front" for text and dimensions over hatch patterns. Use TEXTTOFRONT to handle all annotations at once.

## **Rec: TTF** TEXTTOFRONT

Brings all text, dimension, and leader annotations in the drawing to the front of all other objects in one step.

*Run before plotting sheets to ensure no hatch or solid fill pattern obscures dimensions or room tags.*

**Rec: BL** **BLEND**

Creates a smooth spline curve between the endpoints of two non-touching curves or lines with G1 or G2 continuity.

*Provides superior aesthetic transitions compared to standard fillets when designing aerodynamic or product contours.*

**EXTRIM** **EXTRIM**

Express Tool that trims all intersecting geometry inside or outside a specified boundary polygon in one click.

*Select cutting boundary, then click on the side (inside or outside) that you want trimmed away instantly.*

**HATCHTOBACK** **HATCHTOBACK**

Sends all hatch objects behind all other drawing geometry automatically.

*The counterpart to TEXTTOFRONT. Instantly fixes drawings where hatching covers walls, lines, or notes.*

**Rec: QS** **QSELECT**

Quick Select: builds a filtered selection set based on object type, color, layer, linetype, or attribute value.

*Select all circles with radius > 5, or all text on a specific layer throughout the entire drawing in seconds.*

**FI** **FILTER**

Opens the Object Selection Filter dialog to build complex multi-criteria boolean search filters.

*Create and name reusable filters (e.g., "All Red Lines on Layer WALLS") for complex drawings.*

**DLI DIMLINEAR**

Creates a horizontal or vertical linear dimension.

*Press Enter at the first prompt to select the object directly instead of picking two extension origin points.*

**DAL DIMALIGNED**

Creates a linear dimension parallel to the angle of the specified extension points.

*Use for angled walls, rafters, and diagonal distances that are not orthogonal.*

**DAN DIMANGULAR**

Measures and displays the angle between two selected lines, vertices, or circle sectors.

*Pick two lines to dimension the acute or obtuse angle between them depending on where you place the dimension text.*

**DRA DIMRADIUS**

Creates a radius dimension for a circle or arc with an optional center mark.

*OCS prefixes the text with the standard R symbol. Drag cursor inside or outside curve for placement.*

**DDI DIMDIAMETER**

Creates a diameter dimension for a circle or arc with the diameter symbol (Ø).

*Standard drafting practice: use diameter (DDI) for full circles and radius (DRA) for partial arcs.*

**DAR DIMARC**

Measures and displays the arc length of an arc segment or polyline arc.

*Displays an arc length symbol above or beside the dimension text to distinguish from linear chord distances.*

**DCO DIMCONTINUE**

Creates a chained dimension that continues from the second extension line of the previous dimension.

*Saves enormous time drafting continuous architectural wall strings and column grid dimensions.*

**DBA DIMBASELINE**

Creates a series of stacked dimensions that start from a common baseline extension line.

*Line spacing between stacked baselines is governed by the DIMDLI dimension style variable.*

**DOR DIMORDINATE**

Creates ordinate dimensions showing X or Y coordinate distances from an established datum origin point.

*Commonly used in mechanical hole tables and CNC manufacturing coordinates to prevent tolerance stack-up.*

**D DIMSTYLE**

Opens the Dimension Style Manager to create, modify, compare, and set active dimension styles.

Use Annotative dimension styles so dimensions automatically scale themselves to fit different paper space viewports.

#### **DOV** **DIMOVERRIDE**

Clears or applies temporary overrides to system variables for selected dimensions without modifying the style.

*Useful for quick one-off adjustments like suppressing a single extension line.*

#### **DED** **DIMEDIT**

Edits dimension text, rotation angle, and oblique extension line angles.

*Type O (Oblique) to angle extension lines when dimensioning isometric drawings.*

#### **Rec: QD** **QDIM**

Quickly creates a complete chain or baseline string of dimensions from a group of selected objects in one action.

*Select multiple vertices or lines across a floorplan; QDIM dimensions all selected points in a single click.*

#### **Rec: DRE** **DIMREASSOCIATE**

Associates or reassociates selected dimensions to geometry points so they update automatically when geometry changes.

*Fixes yellow exclamation mark badges indicating orphaned or disassociated dimensions.*

#### **Rec: DSP** **DIMSPACE**

Adjusts the spacing between parallel dimensions or aligns them cleanly.

*Enter Auto or specify a set spacing distance to make dimension strings look publication-grade.*

#### **Rec: DBR** **DIMBREAK**

Breaks or restores dimension lines and extension lines where they cross other dimensions or geometry.

*Prevents drafting clutter where extension lines cross dimension leader lines. Run Auto for automatic clean gaps.*

#### **Rec: CM** **CENTERMARK**

Creates an associative center mark at the center of a circle or arc.

*Associative center marks move and scale dynamically when the underlying circle is moved or resized.*

#### **Rec: CL** **CENTERLINE**

Creates associative centerline geometry between two non-parallel or parallel lines.

*Automatically calculates the midpoint axis between two lines or wall faces and maintains associativity.*

**MT MTEXT**

Creates a multiline text object with full paragraph formatting, columns, bullets, and fractions (also alias T).

Type %%c for diameter symbol (Ø), %%d for degree symbol (°), and %%p for plus/minus (±) in the in-place editor.

**DT TEXT**

Creates single-line text objects where each line is an independent entity.

Use Justify (J) options (MC for Middle Center, TC for Top Center) for clean alignment inside room labels or title blocks.

**MLD MLEADER**

Creates a multileader object connecting text or block callouts to geometry with an arrow and landing.

Configure leader arrowhead, landing distance, and default callout block in MLEADERSTYLE (MLS).

**MLE MLEADEREDIT**

Adds leader lines to or removes leader lines from an existing multileader object.

Allows pointing one note callout to multiple parts or locations simultaneously.

**Rec: TB TABLE**

Creates an empty table object in the drawing with rows, columns, formulas, and Excel data links.

Link directly to an external Excel spreadsheet (.xlsx) so data updates live when the spreadsheet changes.

**Rec: TEX TABLEEXPORT**

Exports data from a drawing table object to a comma-delimited (.csv) file.

Export bills of materials, schedules, and revision histories into Excel or database systems.

**Rec: FD FIELD**

Inserts dynamic text with a field that updates automatically as object properties or drawing metadata change.

Embed dynamic sheet numbers, drawing save dates, viewport scales, or closed polyline calculated areas.

**Rec: UF UPDATEFIELD**

Manually recalculates and updates fields in selected objects to reflect latest drawing values.

Run before issuing prints if FIELDEVAL variable is not set to update upon save or plot.

**Rec: FN FIND**

Finds, replaces, selects, or zooms to text strings, attributes, dimensions, and table cells.

Supports wildcards (\* and ?) and regular expressions. Great for global room numbering or revision title updates.

**SP SPELL**

Checks spelling of text, dimensions, block attributes, and table cells in the drawing.

Choose custom engineering/architectural dictionaries to avoid false alarms on industry acronyms.

## **ST** STYLE

Creates, modifies, or sets named text styles (fonts, height, oblique angle, true-type vs SHX).

*Leave text height at 0 in the text style dialog to allow custom heights per dimension style or annotation note.*

## **Rec: T2M** TXT2MTXT

Converts multiple single-line text (TEXT/DTEXT) objects into a single cohesive MTEXT paragraph block.

*Combines fragmented legacy notes imported from older drawings into an easily editable paragraph.*

## **Rec: TC** TCOUNT

Sequentially numbers text objects in X order, Y order, or selection order with prefixes and suffixes.

*Express Tool for rapid room numbering, parking bay counts, or revision marker sequences.*

## **DX** DATAEXTRACTION

Extracts geometric data, block attribute values, and drawing metadata into a dynamic table or external Excel file.

*Used to generate live doors, windows, and equipment schedules that dynamically update when drawing geometry changes.*

## **DL** DATALINK

Establishes a live bidirectional data link between an Excel spreadsheet (.xlsx) and an OCS table.

*Update linked tables after changing Excel data using the DATALINKUPDATE command.*

## **TCOUNT** TCOUNT

Express Tool that automatically numbers or re-numbers selected text entities sequentially (1, 2, 3...).

*Supports prefix, suffix, overwrite, and sort by X, Y, or selection order. Perfect for parking stalls and room numbers.*

## **TEXTMASK** TEXTMASK

Express Tool that creates a background mask behind single-line or multi-line text to hide underlying linework.

*Use UNMASK to remove the mask if text is relocated.*

## **TEXTTOFRONT** TEXTTOFRONT

Brings all text, dimensions, and leaders to the front of all other drawing objects across the drawing.

*Select "Both" to ensure hatching and filled solids never obscure annotations and dimensions.*

## **FIND** FIND

Finds and replaces text across single-line text, MText, block attributes, dimension annotations, and table cells.

*Check "Match case" or "Find whole words only". Supports project-wide replace in seconds.*

**LA LAYER**

Opens the Layer Properties Manager palette to manage layer colors, linetypes, lineweights, transparency, and viewport overrides.

*Use layer filters (group or property filters) to manage complex drawings with hundreds of AIA/ISO CAD standard layers.*

**PR PROPERTIES**

Opens the Properties palette (Ctrl+1) to inspect and modify all geometric and styling parameters of selected objects.

*Aliases CH, MO also open this palette. Use the Quick Select (funnel icon) button at top-right for precision filtering.*

**MA MATCHPROP**

Applies the properties (layer, color, linetype, text style, dimension style) of a selected source object to other target objects.

*Type S (Settings) after selecting source to toggle exactly which properties to copy (e.g. only layer without changing hatch...*

**Rec: LI LAYISO**

Isolates the layers of selected objects by turning off, freezing, or locking/fading all other layers in the drawing.

*Run S (Settings) to choose whether non-selected layers are completely turned Off or merely Locked and Faded.*

**Rec: LU LAYUNISO**

Restores all layers that were turned off, frozen, or locked by the most recent LAYISO command.

*Quickly returns your workspace to its full visible state after isolated editing.*

**Rec: L0 LAYOFF**

Turns off the layer of any object you click directly in the viewport.

*Clicking any element on the screen turns its layer off without needing to hunt for the name in the layer dropdown.*

**Rec: LON LAYON**

Turns on all layers in the drawing.

*Instantly illuminates every layer that was turned off by LAYOFF or manual layer manager settings.*

**Rec: LF LAYFRZ**

Freezes the layers of selected objects, suppressing them from regenerating and improving performance.

*In paper space viewports, freezes objects in the active viewport only (VP Freeze) without affecting other sheets.*

**Rec: LTW LAYTHW**

Thaws all frozen layers in the drawing.

*Global counterpart to LAYFRZ.*

**Rec: LWK** **LAYWALK**

Opens an interactive layer inspector dialog to quickly walk through and view objects on individual layers.

*The ultimate tool for CAD quality assurance and checking what rogue entities sit on unexpected layers.*

**Rec: LM** **LAYMCH**

Changes the layer of selected objects to match the layer of a destination object.

*Select objects to move, type LAYMCH, and click an object already on the desired target layer.*

**Rec: LC** **LAYCUR**

Changes the layer property of selected objects to the currently active layer.

*Quickly stamps selected stray objects onto whatever layer you are actively drafting on.*

**Rec: LMRG** **LAYMRG**

Merges all objects on selected layers into a single target layer and deletes the empty source layers.

*Safely eliminates duplicate layers (e.g. merging "A-WALL-EXT" and "A-WALL") without manually moving objects.*

**Rec: LDEL** **LAYDEL**

Deletes a layer and permanently erases all objects residing on it, even if referenced in blocks.

*Use with extreme caution. Bypasses standard OCS protections on non-purgeable layers.*

**LTS** **LTSCALE**

Sets the global linetype scale factor for all dotted, dashed, and phantom lines in the drawing.

*Pair with PSLTSCALE=1 so dashed lines appear at identical physical spacing across viewports of different scales.*

**SETBYLAYER** **SETBYLAYER**

Changes the property overrides of selected objects to ByLayer (color, linetype, lineweight, material).

*Essential when receiving messy contractor drawings with hard-coded colors that defy layer management.*

**LAYDEL** **LAYDEL**

Deletes an un-purgeable layer and all objects residing on it, even if referenced in blocks.

*Use with extreme caution. Force-removes stubborn layers that standard PURGE cannot delete.*

**LAYMRG** **LAYMRG**

Merges all objects on selected layers into a single target layer and automatically deletes the source layers.

*Consolidates duplicate or redundant contractor layers without losing a single line or block.*

**LAYTRANS** **LAYTRANS**

Translates layer names and standards in the current drawing to match another company or AIA layer standard file.

*Save translation mappings to a .dws or .dwg standard file for recurring project conversion.*

Rec: VPL

## VPLAYER

Sets viewport-specific layer visibility (VP Freeze, VP Thaw, VP Color) without affecting other viewports.

*Essential for showing plumbing annotations in one viewport while hiding them in the electrical layout on the same sheet.*

**Z ZOOM**

Increases or decreases the apparent magnification of the active viewport drawing window.

*Z -> E (Extents) zooms to all drawn geometry. Z -> W (Window) zooms to a box. Z -> P returns to Previous view.*

**P PAN**

Shifts the display view without changing the magnification level.

*Most drafters click and drag the middle mouse scroll wheel button directly instead of typing the command.*

**RE REGEN**

Regenerates the entire drawing to recompute screen coordinates, smoothing faceted circles and updating linetypes.

*When circles look like segmented octagons after zooming in, running RE instantly restores smooth curved display.*

**REA REGENALL**

Regenerates the entire drawing across all open tiled viewports and paper space viewports.

*Essential on multi-viewport sheets when zooming changes linetype scale appearances across viewports.*

**V VIEW**

Saves and restores named views, camera angles, visual styles, and background overrides.

*Save named model views of key project details to jump directly to them or place them into paper space layouts.*

**Rec: VP VPORTS**

Creates multiple split viewports in model space or manages paper space layout viewport configurations.

*Allows splitting model space into side-by-side plan and section views for simultaneous drafting.*

**MV MVIEW**

Creates and controls rectangular, polygonal, or object-based layout viewports in Paper Space.

*Always lock viewport display (Display Locked = Yes) in the Properties palette once scale is set to prevent accidental scrolling.*

**Rec: PLN PLAN**

Displays an orthogonal plan view of the drawing relative to the Current UCS or World Coordinate System (WCS).

*Type PLAN -> C to rotate the screen so your custom angled UCS sits flat horizontally on your monitor.*

**3DO 3DORBIT**

Rotates the 3D view interactively in 3D space around a target center point.

*Hold Shift + Middle Mouse Wheel to orbit dynamically in 3D without invoking the command prompt.*

**Rec: VB VIEWBASE**

Generates 2D associative orthographic drawing views (Top, Front, Isometric) in paper space directly from a 3D solid model or Inventor file.

*Automates production drawings. Moving or modifying the 3D model automatically updates the 2D layout views.*

**Rec: VPR** **VIEWPROJ**

Generates one or more projected orthographic or isometric views from an existing VIEWBASE drawing view.

*Drag cursor up, down, left, or diagonally to create corresponding elevations and isometric callouts.*

**Rec: VS** **VIEWSECTION**

Generates an associative section view (Full, Half, Offset, Aligned) from a VIEWBASE parent view.

*Draw cutting plane across parent view; OCS automatically hatches sliced solids and displays section callout letters.*

**V** **VIEW**

Saves and restores named Model and Paper Space views, cameras, 3D visual styles, and background settings.

*Named views can be directly dragged from the Sheet Set Manager into layout viewports with auto-scale.*

**VS** **VISUALSTYLES**

Creates and modifies 3D visual styles (2D Wireframe, Conceptual, Hidden, Realistic, Shaded, X-Ray).

*Switch display mode to inspect 3D solid intersections and realistic renderings.*

**Rec: VP** **VPORTS**

Divides the model space display into multiple tiled viewing windows (e.g., Plan, Front, Isometric simultaneously).

*In layout paper space, creates new floating viewport windows for scale drawing.*

**B BLOCK**

Defines a block definition from selected objects in the current drawing.

*Always draw block geometry on Layer 0 with Color and Linetype set to ByBlock so it inherits properties of the inserting layer.*

**I INSERT**

Inserts a named block or drawing file into the current drawing via the Blocks palette or classic dialog.

*Use CLASSICINSERT for the legacy modal dialog box if you prefer simple text inputs without opening the palette.*

**W WBLOCK**

Writes selected objects or an internal block to a separate external DWG drawing file on disk (Write Block).

*The gold standard way to export custom components, details, or clean symbols into your team standard block library.*

**BE BEDIT**

Opens the Block Editor environment to define dynamic parameters, lookup tables, and actions (stretch, flip, visibility).

*Close with BCLOSE (BC) and save changes with BSAVE (BS).*

**ATT ATTDEF**

Creates an attribute definition for storing dynamic text, part numbers, or sheet information inside blocks.

*Set Mode to "Preset" for default text, or "Constant" for unchangeable metadata. Essential for smart title blocks.*

**Rec: BTM BATTMAN**

Opens the Block Attribute Manager to edit attribute properties, order, prompt text, and visibility across definitions.

*Click "Sync" after editing to propagate attribute changes to all pre-existing inserted block instances in the drawing.*

**Rec: ASY ATTSYNC**

Updates and synchronizes block instances with new, reordered, or deleted attribute definitions.

*Run when you added a new attribute inside BEDIT and the older blocks on the drawing do not show the new field.*

**XR XREF**

Opens the External References palette to attach, reload, unload, detach, or bind external DWG, PDF, DWF, and image files.

*Choose "Overlay" rather than "Attachment" type for background architectural plans to avoid circular nesting loops.*

**XA XATTACH**

Directly opens the file browser to select and attach an external DWG drawing as an Xref.

*Faster than opening the Xref manager when quickly bringing in a consultant drawing.*

**XC XCLIP**

Clips the visible boundary of an external reference (Xref) or block using a rectangular or polygonal border.

*Allows isolating just one tenant suite or building wing from a massive master site drawing without exploding.*

**Rec: PDI** **PDFIMPORT**

Imports vector geometry, TrueType text fonts, and raster images from a vector PDF file directly into OCS DWG entities.

*Transforms dead vector PDFs back into live editable lines, arcs, layers, and polylines.*

**Rec: BCNT** **BCOUNT**

Express tool that counts instances of every block in the drawing and lists the totals in the OCS text command window.

*Ideal for quick bills of materials, door counts, and plumbing fixture tallies. Press F2 to expand text window.*

**Rec: CNT** **COUNT**

Modern interactive Count palette that counts instances of blocks, highlights duplicates, and inserts live schedule tables.

*Flags overlapping or mirrored blocks automatically and creates live updating table schedules.*

**ATT** **ATTDEF**

Defines an attribute template (Tag, Prompt, Default value) to store text data within block definitions.

*Essential for smart title blocks, room tags, and equipment labels that prompt the user upon block insertion.*

**BATTMAN** **BATTMAN**

Block Attribute Manager: edits attribute properties, prompt order, visibility, and syncs changes to all existing block instances.

*Use BATTMAN to reorder attributes or change layer/color without exploding or recreating the block.*

**ATTSYNC** **ATTSYNC**

Updates all existing block references with new and revised attribute definitions.

*Always run ATTSYNC after editing a block in BEDIT if new attributes are not showing up in existing instances.*

**EATTEXT** **EATTEXT**

Enhanced attribute data extraction wizard for exporting block attributes and object counts.

*Predecessor to DATAEXTRACTION; still supported out-of-the-box in all standard OCS installations.*

**NCOPY** **NCOPY**

Copies nested objects out of an Xref, DWF, DGN, or Block directly into the current drawing without exploding.

*Click on any line inside an external reference to extract and use it directly in your drawing.*

**XOPEN** **XOPEN**

Select an externally referenced drawing (Xref) and opens its source DWG file directly in a new tab.

*Much faster than navigating through Windows file explorer to find the referenced drawing path.*

**DI DIST**

Measures the distance, angle, and delta X/Y/Z coordinate offsets between two points.

*Use Shift + Right-click to trigger precision object snap overrides (Node, Perpendicular, Midpoint) for measurement.*

**ID ID**

Displays the precise World Coordinate System (X, Y, Z) coordinates of a selected point.

*Quick check to ensure that objects are at Z=0.0000 elevation in 2D drawings.*

**AA AREA**

Calculates the area and perimeter of objects or defined boundary spaces.

*Type O (Object) to click a closed polyline, circle, or hatch. Use Add (A) and Subtract (S) modes for calculating net room areas.*

**LI LIST**

Displays complete database property data for selected objects in the OCS text window (also alias LS).

*Displays exact polyline length, vertex coordinates, layer, handle, linetype, and true 3D normal vector.*

**MEA MEASUREGEOM**

Interactive measurement suite with Quick Mode to display dimensions, distances, and angles dynamically as cursor hovers.

*Quick mode measures room dimensions and clear spans automatically in real-time as you move the cursor inside geometry.*

**Rec: MP MASSPROP**

Calculates the mass, volume, bounding box, center of gravity, moments of inertia, and radii of gyration for 3D solids and regions.

*Output can be saved to an external text data file (.mpr) for structural or mechanical engineering calculations.*

**Rec: TM TIME**

Displays drawing creation date, last saved date, total elapsed editing time, and session timer.

*Check true billable hours and drafting time spent on a project drawing.*

**DX DATAEXTRACTION**

Extracts drawing data and block attributes into a live table or an external file (Excel, Access, CSV).

*Wizard creates automated equipment schedules, door/window schedules, and cable pull sheets from drawing blocks.*

**MASSPROP MASSPROP**

Calculates the mass, volume, bounding box, center of gravity (centroid), and moments of inertia for 3D solids or regions.

*Can write full engineering physical properties analysis to an external .mpr text file.*

**EXT EXTRUDE**

Extrudes 2D closed boundaries or curves along a normal or direction path to create 3D solids or surfaces.

*Use T (Taper angle) option to create draft angles for molded or cast parts.*

**Rec: PP PRESSPULL**

Dynamically pulls or presses bounded areas and faces to extrude or hollow out 3D solid geometry.

*The most intuitive 3D command in OCS: clicking inside any closed boundary immediately adds or subtracts volume.*

**REV REVOLVE**

Creates a 3D solid or surface by sweeping a 2D profile around an axis.

*Commonly used for shafts, bottles, domes, spherical vessels, and circular architectural columns.*

**Rec: LFT LOFT**

Creates a 3D solid or surface in the space between several cross-section profiles.

*Use guide curves or paths to govern transitions between different geometric cross-sections (e.g. circle to square).*

**Rec: SWP SWEEP**

Creates a 3D solid or surface by sweeping an open or closed 2D profile along an open or closed 2D/3D path curve.

*Ideal for pipes, moldings, handrails, conduits, and wiring harnesses.*

**UNI UNION**

Combines two or more 3D solids, surfaces, or 2D regions into a single composite solid.

*Part of the essential 3D CSG boolean modeling toolkit alongside SUBTRACT and INTERSECT.*

**SU SUBTRACT**

Subtracts the volume of one set of 3D solids or regions from another.

*First select objects to keep and press Enter; then select objects to subtract and press Enter.*

**IN INTERSECT**

Creates a composite solid, surface, or region from the intersecting volume of two or more overlapping entities.

*Removes all non-overlapping material, leaving only shared interior volume.*

**BOX BOX**

Creates a 3D solid box primitive.

*Type C (Cube) to ensure all three dimensions (Length, Width, Height) remain equal.*

**CYL CYLINDER**

Creates a 3D solid cylinder primitive.

Specify base center, radius, and height, or use the Elliptical sub-option for oval cylinders.

### **FSHOT** **FLATSHOT**

Creates a 2D flat drawing representation of all 3D solid objects in the current view plane.

*Provides a clean 2D isometric or elevation block with foreground and obscured lines on separate layers.*

### **Rec: SPLN** **SECTIONPLANE**

Creates a section object that acts as a cutting plane through 3D solids, surfaces, and point clouds.

*Toggle Live Sectioning on to see inside buildings or mechanical assemblies in real time.*

### **STEPEXPORT** **STEPEXPORT**

Exports 3D solids, surfaces, and geometry from OCS into ISO 10303 standard STEP (.stp, .step) files for CAM/machining.

*The official out-of-the-box standard OCS command for STEP export (OCS registers EXPORTSTEP/STPOUT).*

### **STLOUT** **STLOUT**

Exports selected 3D solids to binary or ASCII Stereolithography (.stl) mesh files for 3D printing and toolpath CAM.

*Select only watertight 3D solids; standard OCS validates the manifold mesh before exporting.*

### **SEC** **SECTION**

Uses the intersection of a plane and 3D solids to generate a 2D cross-sectional Region object.

*Great for obtaining exact 2D perimeter outlines through complex 3D assemblies at any cut angle.*

### **FSHOT** **FLATSHOT**

Creates a 2D flattened vector representation of all 3D solids based on the current viewing perspective.

*Generates clean 2D elevations and plan views with optional hidden line distinction.*

**OP OPTIONS**

Opens the comprehensive OCS Options dialog to configure file paths, display colors, crosshair size, open/save defaults, and plotting.

*Set Crosshair size to 100 on the Display tab for full-screen drafting crosshairs favored by veteran CAD drafters.*

**UN UNITS**

Controls coordinate and angle display formats (Architectural, Decimal, Engineering) and precision.

*Crucial setting: ensure Insertion Scale units match your project standard (e.g. Inches vs Millimeters) to avoid 25.4x scaling...*

**Rec: RI REINIT**

Reinitializes the I/O ports and reloads the ocad.pgp command aliases file without restarting OCS.

*Essential workflow: after modifying ocad.pgp, run REINIT, check "PGP File", and click OK to load new shortcuts instantly.*

**Rec: AEC AI\_EDITCUSTFILE**

OCS system command that prompts for and opens customization files (such as ocad.pgp) directly in Notepad.

*Type AI\_EDITCUSTFILE -> enter ocad.pgp. Instantly opens the active user PGP file in Notepad without searching folder paths.*

**Rec: ALED ALIASEDIT**

Express Tools graphical dialog box to view, add, edit, and delete command aliases without manual text file editing.

*Available in full OCS. Modifies ocad.pgp automatically and reinitializes on save.*

**AUDIT AUDIT**

Evaluates the database integrity of a drawing and corrects internal errors and orphaned pointers.

*Choose "Y" at the "Fix any errors detected?" prompt. Run regularly on large, sluggish, or crash-prone drawings.*

**Rec: REC RECOVER**

Repairs and opens a severely damaged or corrupted DWG drawing file that refuses to open normally.

*The first line of defense when a drawing crashes on opening with an internal error.*

**Rec: RECALL RECOVERALL**

Repairs and opens a corrupted drawing file and all of its nested attached external references (Xrefs).

*Use when opening complex multi-sheet plans with dozens of linked consultant drawings.*

**PU PURGE**

Removes unused named items (blocks, layers, linetypes, dimension styles, materials) to reduce file size.

*Check "Purge nested items" and click Purge All. Drastically reduces file size and speeds up save and transmission times.*

**QC QUICKCALC**

Opens the QuickCalc interactive scientific and geometric calculator palette (Ctrl+8).

*Directly reads distances, coordinates, and angles from drawing objects and converts units (e.g. feet to meters).*

**Rec: CUI CUI**

Opens the Customize User Interface editor to customize ribbon tabs, toolbars, mouse buttons, and keyboard shortcuts.

*Use the Transfer tab to migrate custom toolbars and workspaces to newer OCS version releases.*

**AP APPLOAD**

Loads and unloads AutoLISP (.lsp), ObjectARX (.arx), and .NET application routines into OCS.

*Drag routines into the "Startup Suite" briefcase to ensure they load automatically every time OCS starts.*

**SCR SCRIPT**

Executes a sequential command script (.scr) file for batch automated tasks.

*Used with batch utility tools like ScriptPro to process hundreds of DWGs overnight (e.g. Purge, Audit, Save).*

**Rec: SVM SYSVARMONITOR**

Monitors critical system variables (FILEDIA, PICKFIRST, MIRRTEXT, EXPERT) and alerts when non-standard values change.

*Saves hours of frustration when FILEDIA flips to 0 and file open dialog boxes stop appearing.*

**Rec: DCMPCOMPARE**

Visually compares two revisions of a drawing and color-codes added (green), deleted (red), and unchanged objects.

*Modern OCS tool for instant design revision tracking and verifying consultant submittals.*

**Rec: PUBPUBLISH**

Batch plots and publishes multi-sheet drawing sets to PDF, DWF, or physical plotters.

*Save sheet list descriptions (.dsd) so you can reprint an entire 50-sheet permit set in a single background click.*

**Rec: PSPAGESETUP**

Configures paper sheet sizes, plot device drivers, CTB/STB pen tables, and orientation for each layout.

*Apply named page setups across your Sheet Set Manager for consistent company standard deliverables.*

**Rec: ETRETRANSMIT**

Packages a drawing and all its associated external references, fonts, plot styles (CTB), and images into a single zip file.

*The standard protocol before emailing drawings to clients or contractors to ensure no missing Xrefs or fonts.*

**SSMSHEETSET**

Opens the Sheet Set Manager palette (Ctrl+4) to manage multi-sheet drawing projects, automated callout numbers, and sheet lists.

*Automates project-wide sheet numbering, cover sheet indices, and automated title block revisions.*

#### **DWGPROPS** **DWGPROPS**

Sets and displays file properties (Title, Subject, Author, Keywords, Custom fields) of the current drawing.

*Custom fields can be dynamically referenced by fields in title blocks and Sheet Sets.*

#### **GEO** **GEOGRAPHICLOCATION**

Assigns real-world GIS coordinates and loads live aerial satellite maps directly under your CAD geometry.

*Enables accurate civil, site layout, and sun-study simulations aligned with real geographic coordinates.*

#### **Rec: MUL** **MULTIPLE**

Repeats the next specified command automatically without needing to press Spacebar or Enter between repetitions.

*Type MULTIPLE -> CIRCLE; now draw circle after circle continuously. Press Esc when finished.*

#### **SAVEALL** **SAVEALL**

Saves all currently open drawing windows and tabs simultaneously without closing them.

*Silently runs QSAVE on all named files; prompts for file path only on newly created unsaved drawings.*

#### **CLOSEALL** **CLOSEALL**

Closes all open drawings, prompting to save changes on any modified documents.

*Use when finishing a project session to close all tabs quickly instead of clicking each X.*

#### **CLOSEALLOTHER** **CLOSEALLOTHER**

Closes all open drawing tabs except the currently active one.

*Cleans up cluttered tab bars when focusing on a single specific drawing.*

#### **Rec: AUD** **AUDIT**

Evaluates the integrity of the drawing database and corrects internal software errors and corrupted pointers.

*When asked "Fix any errors detected? [Yes/No]", type Y. Essential for resolving sluggish or crashing files.*

#### **AP** **APPLOAD**

Loads AutoLISP (.lsp), ObjectARX (.arx), VBA, and .NET applications into OCS.

*Add routinely used .lsp scripts to the "Startup Suite" briefcase icon to auto-load every time OCS starts.*

#### **CUI** **CUI**

Customizes User Interface: edits ribbons, toolbars, mouse buttons, keyboard shortcuts, and double-click actions.

*Manage custom enterprise workspaces and transfer ribbon panels between different machines.*

#### **SET** **SETVAR**

Lists or changes the value of OCS system variables directly from the command prompt.

Type ? to inspect the current state and description of all loaded system variables.

Rec: RI

## REINIT

Reinitializes the I/O ports, digitizer, display, and reloads the ocad.pgp file without restarting OCS.

*Check the "PGP File" checkbox to apply any newly added custom shortcuts immediately.*



# OCS System Variables Registry

96 CORE SYSVARS

Variables fully recognized directly at the OCS command prompt (type directly or invoke transparently via 'VARNAME')

Collapse  
Section



**Recognized at the Command Line:** In standard OCS, typing any of these system variables into the command bar displays its current value in angle brackets (e.g. **OSMODE <4133>**) and prompts for a new value. You can also type them inside active commands by prefixing an apostrophe (e.g. **'OSMODE'**).

Search variables (e.g. OSMODE, LTSCALE, FILEDIA)...

All Read/Write Read-Only

All Functional Groups

All 96

Snaps & Cursor 16

Scale & Display 15

Units & Dimensioning 16

Selection & Grips 16

VARIABLE NAME

GROUP

DEFAULT

TYPE /  
ACCESS

DESCRIPTION & DRAFT  
IMPACT

OSMODE

Snaps &  
Cursor

4133

Bitmask  
Read/Write

Sets running Object Snap (Osnap) modes using bit sum (1=Endpoint, 2=Midpoint, 4=Center, 8=Node, 16=Quadrant, 32=Intersection, 64=Insertion, 128=Perpendicular, 256=Tangent, 512=Near Extension, 1024=Apparent, 2048=Extension, 4096=Parallel).  
Tip: Add 16384 to suppress running osnaps without locking your configured bitmask. turns all osnaps off.

SNAPMODE

Snaps &  
Cursor

0 (Off)

Boolean  
Read/Write

Turns Snap mode on (1) or off (0) — locks cursor movement to discrete grid increments.  
Tip: Toggle instantly using F9 shortcut key.

SNAPANG

Snaps &  
Cursor

0.0

Real  
Read/Write

Sets the snap and grid rotation angle for the current viewport relative to the current UCS.  
Tip: Rotate snap grid to align with crosshairs with angled button.

| VARIABLE NAME | GROUP          | DEFAULT      | TYPE / ACCESS             | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                 |
|---------------|----------------|--------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                |              |                           | wings or site property line                                                                                                                                                                |
| SNAPUNIT      | Snaps & Cursor | 0.5, 0.5     | 2D/3D Point<br>Read/Write | Sets the X and Y snap spacing intervals for the current viewport.<br>Tip: Can be set independently for X and Y for non-square locking.                                                     |
| SNAPTYPE      | Snaps & Cursor | 0 (Standard) | Integer<br>Read/Write     | Sets snap style: 0 for standard rectangular grid, 1 for isometric snap.<br>Tip: Set to 1 to activate 2D isometric drafting mode.                                                           |
| SNAPISOPAIR   | Snaps & Cursor | 0 (Left)     | Integer<br>Read/Write     | Controls the current isoplane when SNAPTYPE=1 (0=Left, 1=Top, 2=Right).<br>Tip: Press F5 or Ctrl+E to toggle isoplanes on the fly while drawing.                                           |
| GRIDMODE      | Snaps & Cursor | 0 (Off)      | Boolean<br>Read/Write     | Turns grid display on (1) or off (0) in the current viewport.<br>Tip: Toggle on/off with F7.                                                                                               |
| GRIDUNIT      | Snaps & Cursor | 0.5, 0.5     | 2D/3D Point<br>Read/Write | Specifies the grid X and Y spacing for the active viewport.<br>Tip: Setting to 0,0 automatically synchronizes grid spacing with SNAPUNIT.                                                  |
| GRIDDISPLAY   | Snaps & Cursor | 2            | Bitmask<br>Read/Write     | Controls grid visual behavior: 1=display beyond drawing limits, 2=adaptive grid, 4=subdivision beyond limits.<br>Tip: Set to 2 to prevent excessive dense grid lines when zooming far out. |
|               |                |              |                           |                                                                                                                                                                                            |

| VARIABLE NAME | GROUP          | DEFAULT     | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                      |
|---------------|----------------|-------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CURSORSIZE    | Snaps & Cursor | 5 (Percent) | Integer<br>Read/Write | Sets the size of the crosshairs as a percentage of the screen size (1 to 100).<br><a href="#">Tip: Set to 100 for full-screen crosshairs spanning the entire drawing window (classic OLE style).</a>            |
| PICKBOX       | Snaps & Cursor | 3 (Pixels)  | Integer<br>Read/Write | Sets the object selection target pickbox height in device pixels (0 to 50).<br><a href="#">Tip: Increase on 4K high-resolution displays to make selecting lines effortless.</a>                                 |
| APERTURE      | Snaps & Cursor | 10 (Pixels) | Integer<br>Read/Write | Sets the display size of the Object Snap target aperture box in pixels.<br><a href="#">Tip: Larger aperture catches snap points from further away.</a>                                                          |
| AUTOSNAP      | Snaps & Cursor | 63          | Bitmask<br>Read/Write | Controls Osnap visual modes (1), tooltip (2), magnet (4), polar tracking (8), and extension paths (16/32).<br><a href="#">Tip: Bit 4 (value 4) is the magnet that pulls the cursor directly to snap points.</a> |
| POLARMODE     | Snaps & Cursor | 0           | Bitmask<br>Read/Write | Controls polar and object snap tracking settings (relative vs. absolute and angle).<br><a href="#">Tip: Set to 1 to measure tracking angles relative to selected segments.</a>                                  |
| POLARANG      | Snaps & Cursor | 90.0        | Real<br>Read/Write    | Sets polar tracking angle increment in degrees (e.g., 30, 45, 90).<br><a href="#">Tip: Set to 45 or 30 for isometric/architectural drawing guides.</a>                                                          |
| ORTHOMODE     | Snaps & Cursor | 0 (Off)     | Boolean<br>Read/Write | Constrains cursor movement strictly to horizontal and vertical axes (F8 toggle).                                                                                                                                |

| VARIABLE NAME | GROUP             | DEFAULT | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                                                         |
|---------------|-------------------|---------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                   |         |                       | Tip: Hold Shift temporarily while drawing to invert Ortho mode the fly.                                                                                                                                                                                                                            |
| PICKADD       | Selection & Grips | 2       | Integer<br>Read/Write | <p>Controls whether subsequent object selections replace or add to the current selection set (0=replace, 1=add, 2=replace and retain selection after command ends).</p> <p>Tip: If clicking a new object deselects your previous object, PICKADD was accidentally changed to 0; set back to 2.</p> |
| PICKAUTO      | Selection & Grips | 5       | Integer<br>Read/Write | <p>Controls automatic window selection when clicking or space (window, crossing, fence, lasso).</p> <p>Tip: Set to 1 for classic window/crossing without accidental lasso dragging.</p>                                                                                                            |
| PICKDRAG      | Selection & Grips | 2       | Boolean<br>Read/Write | <p>Controls how selection windows are drawn: 0=toggle clicks, 1=press and drag, 2=auto-detect both.</p> <p>Tip: Set to 0 if you find you accidentally dragging selection windows.</p>                                                                                                              |
| PICKFIRST     | Selection & Grips | 1 (On)  | Boolean<br>Read/Write | <p>Controls whether you can select objects before starting a command (Noun/Verb selection).</p> <p>Tip: If pressing Delete key while running MOVE no longer works on pre-selected objects, set PICKFIRST to 1.</p>                                                                                 |
| GRIPS         | Selection & Grips | 2       | Integer<br>Read/Write | <p>Controls display of selection grips (0=No grips, 1=Grips displayed, 2=Midpoint grips on polyline segments).</p> <p>Tip: Set to 2 to gain midpoint stretch and arc-convert grips menus on polylines.</p>                                                                                         |

| VARIABLE NAME        | GROUP             | DEFAULT      | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                |
|----------------------|-------------------|--------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRIPSIZE             | Selection & Grips | 5 (Pixels)   | Integer<br>Read/Write | Sets the size of the unselected grip box in pixels (1 to 255).<br><b>Tip:</b> Scale up on 4K monitors to make grabbing entity vertices easier.                            |
| GRIPCOLOR            | Selection & Grips | 160 (Blue)   | Integer<br>Read/Write | Controls the color of unselected (cold) grips using the ACI color index 1-255).<br><b>Tip:</b> Change if working on a dark or customized viewport background.             |
| GRIPHOT              | Selection & Grips | 1 (Red)      | Integer<br>Read/Write | Controls the color of selected (hot) active grips.<br><b>Tip:</b> Hot grips are actively stretched or transformed.                                                        |
| GRIPHOVER            | Selection & Grips | 11 (Pink)    | Integer<br>Read/Write | Controls the color of a hovered grip before clicking.<br><b>Tip:</b> Pausing cursor over grips triggers dynamic menu functional grip menus.                               |
| SELECTIONAREA        | Selection & Grips | 1 (On)       | Boolean<br>Read/Write | Controls whether selection windows and crossing windows are shaded with translucent color.<br><b>Tip:</b> Set to 0 for classic transparent wire outline selection frames. |
| SELECTIONAREAOPACITY | Selection & Grips | 25 (Percent) | Integer<br>Read/Write | Sets transparency percentage of selection area tint (0 to 100).<br><b>Tip:</b> Default 25% provides visual contrast without obscuring underlying drawing lines.           |
| SELECTIONPREVIEW     | Selection & Grips | 3            | Bitmask<br>Read/Write | Controls object glow highlighting when cursor hovers over objects (1=vertical command active, 2=horizontal command active).                                               |

| VARIABLE NAME        | GROUP             | DEFAULT    | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                    |
|----------------------|-------------------|------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                   |            |                       | Tip: Set to 0 on massive c<br>maps or complex files to I<br>viewport cursor speed.                                                                                                                            |
| SELECTIONEFFECT      | Selection & Grips | 1 (Glow)   | Boolean<br>Read/Write | Controls visual selection highlight style: 1=solid g line color, 0=classic das bounding highlight.<br>Tip: Classic drafters who the neon glow set this to dashed lines.                                       |
| SELECTIONEFFECTCOLOR | Selection & Grips | 0          | Integer<br>Read/Write | Sets the custom color us for object selection glow highlighting.<br>Tip: Specify color index o to customize selection co                                                                                      |
| WINDOWAREACOLOR      | Selection & Grips | 150 (Blue) | Integer<br>Read/Write | Specifies the color of the standard left-to-right selection window.<br>Tip: Standard window sele only objects completely enclosed within the bound                                                            |
| CROSSINGAREACOLOR    | Selection & Grips | 90 (Green) | Integer<br>Read/Write | Specifies the color of the right-to-left crossing selection window.<br>Tip: Crossing window sele any object touched or cro by the boundary.                                                                   |
| LTSCALE              | Scale & Display   | 1.0        | Real<br>Read/Write    | Sets global linetype scal factor for all dotted, das and patterned lines acro entire drawing.<br>Tip: In Model Space, set LTSCALE to match your intended plot scale factor 48 for 1/4"=1'-0", 100 for 1:100). |
|                      |                   |            |                       |                                                                                                                                                                                                               |

| VARIABLE NAME | GROUP           | DEFAULT     | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                                     |
|---------------|-----------------|-------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CELTSCALE     | Scale & Display | 1.0         | Real<br>Read/Write    | Sets the current entity linetype scale multiplier for newly created objects.<br><b>Tip:</b> Keep at 1.0; adjust LTSCALE or PSLTSCALE instead of setting individual object scales.                                                                                              |
| PSLTSCALE     | Scale & Display | 1 (On)      | Boolean<br>Read/Write | Controls Paper Space linetype scaling: 1=viewports scaled, dashed lines identical on paper regardless of zoom scale; 0=viewports show model scale.<br><b>Tip:</b> Always keep PSLTSCALE and MSLTSCALE=1 for consistent dashes across viewports at 1/8", 1/4", or other scales. |
| MSLTSCALE     | Scale & Display | 1 (On)      | Boolean<br>Read/Write | Scales linetypes displayed in the model tab by the current annotation scale (CANNOSCALE).<br><b>Tip:</b> Set to 1 so dashed linetypes preview accurately in Model Space as you change annotation scale.                                                                        |
| LWDISPLAY     | Scale & Display | 0 (Off)     | Boolean<br>Read/Write | Controls whether real lineweights (thicknesses) are visually displayed in Model and Paper space.<br><b>Tip:</b> Toggle on to verify pen lineweights before plotting, toggle off for crisp pixel-perfect snapping.                                                              |
| LWDEFAULT     | Scale & Display | 25 (0.25mm) | Integer<br>Read/Write | Sets the default lineweight value in hundredths of a millimeter for objects on the Default lineweight.<br><b>Tip:</b> Standard default is 25 (0.25mm).                                                                                                                         |
| DISPSILH      | Scale & Display | 0 (Off)     | Boolean<br>Read/Write | Controls display of 3D solid silhouette outlines when                                                                                                                                                                                                                          |

| VARIABLE NAME | GROUP           | DEFAULT      | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                  |
|---------------|-----------------|--------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                 |              |                       | wireframe and hidden view styles (0=Off, 1=On).<br><a href="#">Tip: Set to 1 to produce clear silhouette outlines for 3D curved surfaces without i mesh tessellation lines.</a>                                             |
| ISOLINES      | Scale & Display | 4            | Integer<br>Read/Write | Specifies number of wire contour lines displayed on curved surfaces of 3D solids (0 to 2047).<br><a href="#">Tip: Increase to 16 or 32 for higher visual surface fidelity on curved solids.</a>                             |
| MAXACTVP      | Scale & Display | 64           | Integer<br>Read/Write | Sets maximum number of layout viewports that can simultaneously be active and regenerated.<br><a href="#">Tip: If some viewports on multi-detail sheet appear to freeze, increase MAXACTVP (up to 256).</a>                 |
| ZOOMFACTOR    | Scale & Display | 60 (Percent) | Integer<br>Read/Write | Controls mouse wheel zoom increment sensitivity (30 to 100). Higher = faster zoom per wheel notch.<br><a href="#">Tip: Set to 40 for finer precision zooming, or 80 for rapid navigation across large sheets and plans.</a> |
| ZOOMWHEEL     | Scale & Display | 0            | Boolean<br>Read/Write | Toggles mouse wheel zoom direction: 0=forward zoom in; 1=backward zooms out; 1=inverts wheel direction.<br><a href="#">Tip: Set to 1 if you are accustomed to GIS, 3ds Max, or Mac scroll wheel direction.</a>              |
|               |                 |              |                       |                                                                                                                                                                                                                             |

| VARIABLE NAME       | GROUP                | DEFAULT     | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                                   |
|---------------------|----------------------|-------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMMANDLINEFADETIME | Scale & Display      | 2 (Seconds) | Integer<br>Read/Write | <p>Sets number of seconds floating semi-transparent command lines remain visible before fading out.</p> <p>Tip: Increase to 5-10 seconds if you want more time to read command prompts and error output.</p>                                                                 |
| REGENMODE           | Scale & Display      | 1 (On)      | Boolean<br>Read/Write | <p>Controls automatic regeneration of drawing when zooming or panning beyond current display bounds.</p> <p>Tip: Keep at 1 so circles and arcs automatically smooth when zooming in.</p>                                                                                     |
| VIEWRES             | Scale & Display      | 1000        | Integer<br>Read/Write | <p>Sets resolution for circle and curve tessellation on screen (1 to 20000).</p> <p>Tip: If circles look like faceted octagons, run VIEWRES, then press F2 to fast zoom, and set to 1000 or 20000.</p>                                                                       |
| HIGHLIGHT           | Scale & Display      | 1 (On)      | Boolean<br>Read/Write | <p>Controls whether selected objects are highlighted on screen (0=no highlight, 1=highlighted).</p> <p>Tip: Keep at 1. If set to 0, objects will appear unselected even when actively part of selection set.</p>                                                             |
| LUNITS              | Units & Dimensioning | 2 (Decimal) | Integer<br>Read/Write | <p>Sets linear coordinate display unit format: 1=Scientific, 2=Decimal, 3=Engineering (feet &amp; decimal inches), 4=Architectural (feet &amp; fractional inches), 5=Fractional.</p> <p>Tip: Architects and interior designers use 4; civil and mechanical engineers use</p> |
|                     |                      |             |                       |                                                                                                                                                                                                                                                                              |

| VARIABLE NAME | GROUP                | DEFAULT               | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                      |
|---------------|----------------------|-----------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LUPREC        | Units & Dimensioning | 4                     | Integer<br>Read/Write | Sets linear units display precision (decimal place fractional denominator).<br>Tip: In Architectural units 1/16" precision; 6 = 1/64" precision.                                                                                |
| AUNITS        | Units & Dimensioning | 0 (Degrees)           | Integer<br>Read/Write | Sets angular units format:<br>0=Decimal degrees,<br>1=Degrees/minutes/seconds<br>2=Gradians, 3=Radians,<br>4=Surveyor units (N 45° E)<br>Tip: Civil survey plans require (Surveyor units) to parse boundary legal descriptions. |
| AUPREC        | Units & Dimensioning | 0                     | Integer<br>Read/Write | Sets angular units display precision decimal places (0 to 8).<br>Tip: Survey drawings typically require 2 to 4 places for seconds precision.                                                                                    |
| ANGBASE       | Units & Dimensioning | 0.0 (East)            | Real<br>Read/Write    | Sets the direction of angles relative to the current User Coordinate System (0=East, 90=North, 180=West, 270=South).<br>Tip: Civil and surveyor drawings frequently set ANGBASE to 0 so 0 degrees points due North.             |
| ANGDIR        | Units & Dimensioning | 0 (Counter-Clockwise) | Boolean<br>Read/Write | Sets angle positive direction:<br>0=Counter-clockwise (mathematical standard)<br>1=Clockwise (survey standard).<br>Tip: Combine ANGBASE=0 and ANGDIR=1 to establish standard survey bearings.                                   |
| INSUNITS      | Units & Dimensioning | 1 (Inches)            | Integer<br>Read/Write | Specifies drawing unit value for automatic scaling when inserting blocks or Xrefs.<br>(0=Unspecified, 1=Inches, 2=Feet, 4=Millimeters, 5=Centimeters, 6=Meters)                                                                 |

| VARIABLE NAME     | GROUP                | DEFAULT | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                                        |
|-------------------|----------------------|---------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                      |         |                       | Tip: Crucial for DesignCenter and Xrefs. If a metric title inserts 25.4x too small, change INSUNITS in both DWGs.                                                                                                                                                                 |
| INSUNITSDEFSOURCE | Units & Dimensioning | 1       | Integer<br>Read/Write | Sets source content unit when INSUNITS is set to (unspecified).<br>Tip: Prevents unwanted unit conversion popups on legacy files.                                                                                                                                                 |
| INSUNITSDEFTARGET | Units & Dimensioning | 1       | Integer<br>Read/Write | Sets target drawing unit when INSUNITS is set to (unspecified).<br>Tip: Set default target drawing units for legacy files.                                                                                                                                                        |
| DIMASSOC          | Units & Dimensioning | 2       | Integer<br>Read/Write | Controls dimension associativity: 0=explode lines/text, 1=non-associative unified dimension, 2=associative (updates automatically when geometry moves or stretches).<br>Tip: Always maintain DIMASSOC=2 so dimensions dynamically track stretching geometry and revision editing. |
| DIMSCALE          | Units & Dimensioning | 1.0     | Real<br>Read/Write    | Sets overall scale factor applied to dimension variables that specify sizes, distances or offsets.<br>Tip: Set to 0 to enable paper space viewport scaling (DIMSCALE=0 calculates scale factor automatically from viewport zoom).                                                 |
| DIMTXT            | Units & Dimensioning | 0.18    | Real<br>Read/Write    | Sets the height of dimension text when the active text does not have a fixed height.<br>Tip: Set text height in your DIMSTYLE rather than alter DIMTXT directly.                                                                                                                  |
|                   |                      |         |                       |                                                                                                                                                                                                                                                                                   |

| VARIABLE NAME     | GROUP                | DEFAULT     | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                                       |
|-------------------|----------------------|-------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DIMLFAC</b>    | Units & Dimensioning | 1.0         | Real<br>Read/Write    | Sets a global linear scale factor multiplier for all measured dimension distances.<br><br>Tip: Use in detail views (e.g. 2x or 0.5x) if dimensioning geometry scaled manually rather than via viewport.                                                                          |
| <b>DIMCLRD</b>    | Units & Dimensioning | 0 (ByBlock) | Integer<br>Read/Write | Assigns color to dimension lines, arrowheads, and leader lines.<br><br>Tip: Set to 256 (ByLayer) for clean, professional layer management.                                                                                                                                       |
| <b>DIMCLRE</b>    | Units & Dimensioning | 0 (ByBlock) | Integer<br>Read/Write | Assigns color to dimension extension lines.<br><br>Tip: Set to 256 (ByLayer) for CAD standards consistency.                                                                                                                                                                      |
| <b>DIMCLRT</b>    | Units & Dimensioning | 0 (ByBlock) | Integer<br>Read/Write | Assigns color to dimension annotation text.<br><br>Tip: Set to 256 (ByLayer) as text color is governed by assigned dimension layer.                                                                                                                                              |
| <b>IMAGEFRAME</b> | Frames & Viewports   | 1           | Integer<br>Read/Write | Controls raster image border frames: 0=frame hidden; 1=frame displayed and plotted; 2=frame displayed but NOT plotted.<br><br>Tip: Setting 2 is optimal: allows you to select, move, and delete images and aerial photos while preventing ugly black border lines from plotting. |
| <b>PDFFRAME</b>   | Frames & Viewports   | 1           | Integer<br>Read/Write | Controls underlay border frame display for attached PDF documents (0=hidden; 1=plotted, 2=visible but not plotted).<br><br>Tip: Set to 2 to work with attached PDF underlays visible in rectangular plot border.                                                                 |
|                   |                      |             |                       |                                                                                                                                                                                                                                                                                  |

| VARIABLE NAME | GROUP              | DEFAULT | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                             |
|---------------|--------------------|---------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DWFFRAME      | Frames & Viewports | 1       | Integer<br>Read/Write | Controls underlay bound frames for attached Auto DWG/DWFX underlays.<br>Tip: Follows identical 0, 1, settings as PDFFRAME and IMAGEFRAME.                                                                                              |
| DGNFRAME      | Frames & Viewports | 1       | Integer<br>Read/Write | Controls underlay frame visibility for attached Bentley MicroStation DGN files.<br>Tip: 0=hidden, 1=plotted, 2=visible not plotted.                                                                                                    |
| WIPEOUTFRAME  | Frames & Viewports | 1       | Integer<br>Read/Write | Controls wipeout bound frames: 0=hidden & not plotted; 1=displayed & plotted; 2=displayed on screen but NOT plotted.<br>Tip: Set to 2 so drafters can see and grip wipeout edges while printing thick perimeter lines.                 |
| XCLIPFRAME    | Frames & Viewports | 0       | Integer<br>Read/Write | Controls clipping bound frame visibility for clipped Xrefs (0=hidden, 1=plotted, 2=displayed not plotted).<br>Tip: Set to 2 to easily adjust clip boundaries without plotting the border box.                                          |
| FRAME         | Frames & Viewports | 3       | Integer<br>Read/Write | Master override variable sets IMAGEFRAME, PDFFRAME, WIPEOUTFRAME and XCLIPFRAME simultaneously in a single command.<br>Tip: Type FRAME -> 2 to set underlay and wipeout border to "visible on screen, suppressed on plot" in one step. |
| VISRETAIN     | Frames & Viewports | 1 (On)  | Boolean<br>Read/Write | Controls Xref layer property retention: 1=keeps layer colors/visibility overrides in host drawing; 0=reset                                                                                                                             |

| VARIABLE NAME | GROUP              | DEFAULT            | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                          |
|---------------|--------------------|--------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                    |                    |                       | external file values every drawing opens.<br><br>Tip: Keep at 1. If layer freeze colors reset every time you open a project DWG, VISF was turned to 0.                                                                                              |
| UCSICON       | Frames & Viewports | 3                  | Bitmask<br>Read/Write | Controls coordinate system icon: 0=Off, 1=On, 2=Plan at Origin (0,0,0), 3=On at Origin.<br><br>Tip: Type UCSICON -> No to dock the coordinate man in the lower-left corner in of floating at 0,0.                                                   |
| UCSVP         | Frames & Viewports | 1 (On)             | Boolean<br>Read/Write | Controls whether viewport UCS remains locked or updates when switching between layout viewport.<br><br>Tip: Allows each layout vi to maintain its own indepe rotation or plan orientation                                                           |
| DELOBJ        | 3D & Shading       | 1                  | Integer<br>Read/Write | Controls whether source profile linework is retained deleted when creating 3 solids (0=retain all, 1=delete profiles/paths, 2=delete source curves).<br><br>Tip: Set to 0 if you want to original 2D sketch paths a running EXTRUDE, REVOLVE SWEEP. |
| FACETRES      | 3D & Shading       | 0.5                | Real<br>Read/Write    | Adjusts smoothness of s 3D solids, curves, and rendered views (0.01 to<br><br>Tip: Set to 5.0 or 10.0 for publication-quality shade exports with silky smooth curved surfaces.                                                                      |
| SURFTYPE      | 3D & Shading       | 6 (Cubic B-spline) | Integer<br>Read/Write | Surface fitting type for smoothed polygon mesh<br>5=Quadratic B-spline, 6=Cubic B-spline, 8=Be surface.                                                                                                                                             |

| VARIABLE NAME    | GROUP         | DEFAULT | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                   |
|------------------|---------------|---------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |               |         |                       | Tip: Used with the EDGECLIP and REVSURF mesh modeling tools.                                                                                                                                                                                                 |
| <b>SURFU</b>     | 3D & Shading  | 6       | Integer<br>Read/Write | Surface density in the M direction for 3D polygon meshes (2 to 200).<br>Tip: Higher density increases mesh smoothness at the expense of drawing memory.                                                                                                      |
| <b>SURFV</b>     | 3D & Shading  | 6       | Integer<br>Read/Write | Surface density in the N direction for 3D polygon meshes (2 to 200).<br>Tip: Controls longitudinal resolution.                                                                                                                                               |
| <b>THICKNESS</b> | 3D & Shading  | 0.0     | Real<br>Read/Write    | Sets default 3D extrusion thickness applied to newly drawn 2D objects (lines, circles, arcs).<br>Tip: If 2D lines suddenly look like tall 3D walls when viewed in 3D orbit, THICKNESS was accidentally set > 0; set back to 0.                               |
| <b>ELEVATION</b> | 3D & Shading  | 0.0     | Real<br>Read/Write    | Sets default Z-elevation for newly drawn entities relative to the current UCS plane.<br>Tip: If geometry refuses to lie in 2D Z=0, check ELEVATION and set back to 0.                                                                                        |
| <b>FILEDIA</b>   | File & System | 1 (On)  | Boolean<br>Read/Write | Controls display of file navigation dialog boxes (1=GUI file picker dialog, 0=command-line prompts only).<br>Tip: The classic OCS panels OPEN, SAVEAS, or XREF suddenly only prompt in the command line without opening a window, type FILEDIA and set to 1. |
| <b>CMDDIA</b>    | File & System | 1 (On)  | Boolean<br>Read/Write | Controls display of command dialog boxes (e.g. PLOT,                                                                                                                                                                                                         |

| VARIABLE NAME | GROUP         | DEFAULT      | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                                                                                         |
|---------------|---------------|--------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |               |              |                       | QLEADER, ATTDEF dialog boxes.<br>Tip: Set to 0 inside LISP scripts to automate plotting; set to 1 for interactive GUI dialogs.                                                                                                                                     |
| CMDECHO       | File & System | 1 (On)       | Boolean<br>Read/Write | Controls whether command prompts and echo output are printed to the command window during script/LISP execution.<br>Tip: Set to 0 in AutoLISP programs to suppress text flicker and maximize execution speed.                                                      |
| MIRRTEXT      | File & System | 0 (Off)      | Boolean<br>Read/Write | Controls how the MIRROR command reflects text: 0=retains text reading direction (left-to-right), 1=mirrors text backward as a mirror image.<br>Tip: Always keep MIRRTEXT=0 so mirrored floor plans or symmetrical parts do not produce backwards unreadable notes. |
| TEXTFILL      | File & System | 1 (Filled)   | Boolean<br>Read/Write | Controls whether TrueType (TTF) fonts are displayed as solid fills or as wireframe outlines.<br>Tip: Set to 0 to outline TrueType fonts for laser cutting, engraving, or CNC routing.                                                                              |
| TEXTQLTY      | File & System | 50 (Percent) | Integer<br>Read/Write | Sets resolution and smoothness of TrueType outlines (0 to 100).<br>Tip: Increase to 100 for sharper vector plotting and profiling.                                                                                                                                 |
| PLINEGEN      | File & System | 1 (On)       | Boolean<br>Read/Write | Controls linetype dash generation around 2D plot vertices: 1=continuous dash pattern throughout line, 0=restarts pattern at each vertex.                                                                                                                           |

| VARIABLE NAME | GROUP         | DEFAULT        | TYPE / ACCESS         | DESCRIPTION & DRAFT IMPACT                                                                                                                                                                   |
|---------------|---------------|----------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |               |                |                       | Tip: Set to 1 so dashed lines around rounded corners can appear solid or broken.                                                                                                             |
| SAVETIME      | File & System | 10 (Minutes)   | Integer<br>Read/Write | Sets automatic backup save interval in minutes (0 to 10 disables autosave).<br>Tip: Recommended 5-10 minutes. Autosave files are written with .sv\$ extension in your Windows %TEMP% folder. |
| STARTUP       | File & System | 3              | Integer<br>Read/Write | Controls startup behavior when launching OCS (0=no drawing, 1=Startup dialog, 2=no drawing, 3=Start to screen).<br>Tip: Set to 0 to bypass the tab and immediately open default template.    |
| DWGNAME       | File & System | Drawing1.dwg   | String<br>Read-Only   | Stores the file name of the currently active drawing document.<br>Tip: Referenced dynamically in Fields inside title blocks to auto-print sheet file name.                                   |
| DWGPREFIX     | File & System | C:\Drawings\   | String<br>Read-Only   | Stores the drive and directory folder path of the active drawing document.<br>Tip: Used in title block sheet metadata fields and AutoCAD directory routing.                                  |
| LOGINNAME     | File & System | (Windows User) | String<br>Read-Only   | Stores the Windows or Autodesk account user login name.<br>Tip: Used in title blocks and stamps to track "Drawn By" approval records automatically.                                          |
| AREA          | File & System | 0.0            | Real<br>Read-Only     | Stores the last calculated surface or perimeter area using AREA, MEASUREGEOM, MASSPROP.<br>Tip: Can be linked into dynamic text fields for live square-                                      |

| VARIABLE NAME | GROUP         | DEFAULT       | TYPE / ACCESS       | DESCRIPTION & DRAFT IMPACT                                                                                                                          |
|---------------|---------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|               |               |               |                     | footage annotations.                                                                                                                                |
| PERIMETER     | File & System | 0.0           | Real<br>Read-Only   | Stores the last calculated perimeter or boundary length computed by inquiry commands.<br>Tip: Referenced in automatic takeoff calculations.         |
| CDATE         | File & System | 20260912.0558 | Real<br>Read-Only   | Stores current calendar date and time in decimal format (YYYYMMDD.HHMMSS).<br>Tip: Used in plot stamps, block dates, and revision logging routines. |
| PRODUCT       | File & System | OCS           | String<br>Read-Only | Returns the official product name of the active CAD platform.<br>Tip: Used by plugins to differentiate the OCS engine from full OCS or OCS.         |

Showing 96 of 96 standard OCS system variables.

All variables accept command-line entry and transparent execution ('VAR).

# OCS Extended & Engine-Registered Command Registry

Completing all 443 documented registry entries: In addition to the standard 206 drafting commands and 96 system variables, the sections below account for duplicate alias registrations, shell/clipboard primitives, and recognized 3D sub-commands.

 Search extended commands...

- All 3 Extended Sections (141)
-  Duplicate Registered Aliases (74)
-  Application & Clipboard Meta (39)
-  3D & Mesh (28)



## Duplicate Aliases Registered as Independent Commands 74 Entries

In OCS's Rust architecture, both the full command name (e.g. 3DORBIT) and its shorthand aliases (e.g. 30) are submitted as distinct `inventory::submit!` commands.

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY               | REGISTERED SOURCE MODULE | REGISTRATION<br>CONTEXT &<br>PURPOSE                                                          |
|----------------|-----------------------------|------------------------|--------------------------|-----------------------------------------------------------------------------------------------|
| 30             | *3DORBIT                    | View &<br>Navigation   | src/app/commands/mod.rs  | Short two-letter orbit shorthand registered as first-class CadCommand token alongside 3DORBIT |
| LA             | *LAYER                      | Layers &<br>Properties | src/app/commands/mod.rs  | LAYER primary alias registered directly in command dispatcher                                 |
| LAYERS         | *LAYER                      | Layers &<br>Properties | src/app/commands/mod.rs  | Common drafted pluralization typo registered as direct command                                |
| PR             | *PROPERTIES                 | Layers &<br>Properties | src/app/commands/mod.rs  | Standard 2-letter property inspector shortcut registered as command                           |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY               | REGISTERED SOURCE MODULE         | REGISTRATION<br>CONTEXT &<br>PURPOSE                                        |
|----------------|-----------------------------|------------------------|----------------------------------|-----------------------------------------------------------------------------|
| <b>PROPS</b>   | <b>*PROPERTIES</b>          | Layers &<br>Properties | src/app/commands/mod.rs          | Colloquial<br>abbreviation<br>registered as<br>direct command               |
| <b>CH</b>      | <b>*PROPERTIES</b>          | Layers &<br>Properties | src/app/commands/mod.rs          | Legacy<br>CHANGE /<br>PROPERTIES<br>shortcut<br>registered<br>independently |
| <b>MO</b>      | <b>*PROPERTIES</b>          | Layers &<br>Properties | src/app/commands/mod.rs          | Legacy MODIFY<br>shortcut<br>registered<br>independently                    |
| <b>DDIM</b>    | <b>*DIMSTYLE</b>            | Dimension              | src/app/commands/mod.rs          | Legacy dialog<br>command<br>registered<br>alongside<br>DIMSTYLE             |
| <b>D</b>       | <b>*DIMSTYLE</b>            | Dimension              | src/app/commands/mod.rs          | Single key<br>dimension style<br>shortcut<br>registered as<br>token         |
| <b>MEA</b>     | <b>*MEASUREGEOM</b>         | Inquiry &<br>Measure   | src/app/commands/mod.rs          | MEASUREGEOM<br>alias registered<br>as independent<br>dispatch entry         |
| <b>DCE</b>     | <b>*DIMCENTER</b>           | Dimension              | src/app/commands/mod.rs          | Center mark<br>shortcut<br>registered as<br>separate<br>command entry       |
| <b>DSPACE</b>  | <b>*DIMSPACE</b>            | Dimension              | src/modules/annotate/dimspace.rs | Dimension<br>space<br>abbreviation<br>registered<br>directly                |
| <b>QC</b>      | <b>*QUICKCALC</b>           | System &<br>Utilities  | src/app/commands/mod.rs          | Quick calculator<br>registered as<br>individual<br>command                  |
| <b>XR</b>      | <b>*XREF</b>                | Blocks &<br>References | src/app/commands/mod.rs          | External<br>reference                                                       |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY               | REGISTERED SOURCE MODULE | REGISTRATION<br>CONTEXT &<br>PURPOSE                              |
|----------------|-----------------------------|------------------------|--------------------------|-------------------------------------------------------------------|
|                |                             |                        |                          | manager alias<br>registered<br>directly                           |
| <b>XA</b>      | <b>*XATTACH</b>             | Blocks &<br>References | src/app/commands/mod.rs  | Xref attach<br>dialog alias<br>registered as<br>command           |
| <b>XB</b>      | <b>*XBIND</b>               | Blocks &<br>References | src/app/commands/mod.rs  | Xref dependent<br>symbols binding<br>alias registered<br>directly |
| <b>WB</b>      | <b>*WBLOCK</b>              | Blocks &<br>References | src/app/commands/mod.rs  | Write Block<br>shortcut<br>registered as<br>distinct<br>command   |
| <b>TS</b>      | <b>*STYLE</b>               | Annotation             | src/app/commands/mod.rs  | Text style<br>shortcut<br>registered as<br>separate<br>command    |
| <b>P</b>       | <b>*PAN</b>                 | View &<br>Navigation   | src/app/commands/mod.rs  | Pan shortcut<br>registered as<br>separate entry                   |
| <b>ZS</b>      | <b>*ZOOM</b>                | View &<br>Navigation   | src/app/commands/mod.rs  | Zoom window /<br>scale shorthand<br>registered<br>directly        |
| <b>QS</b>      | <b>*QSAVE</b>               | System &<br>Utilities  | src/app/commands/mod.rs  | Quick Save<br>shortcut<br>registered as<br>distinct token         |
| <b>SA</b>      | <b>*SAVEAS</b>              | System &<br>Utilities  | src/app/commands/mod.rs  | Save As<br>shortcut<br>registered as<br>separate<br>command       |
| <b>DE</b>      | <b>*DDEDIT</b>              | Annotation             | src/app/commands/mod.rs  | Text edit<br>shortcut<br>registered<br>directly                   |
| <b>HB</b>      | <b>*HATCHTOBACK</b>         | Modify                 | src/app/commands/mod.rs  | Hatch to back<br>shortcut                                         |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY | REGISTERED SOURCE MODULE         | REGISTRATION<br>CONTEXT &<br>PURPOSE                     |
|----------------|-----------------------------|----------|----------------------------------|----------------------------------------------------------|
|                |                             |          |                                  | registered<br>directly                                   |
| <b>BLE</b>     | <b>*BLEND</b>               | Modify   | src/modules/draw/modify/blend.rs | Blend curve<br>shorthand<br>registered as<br>command     |
| <b>L</b>       | <b>*LINE</b>                | Draw     | src/modules/draw/draw/line.rs    | LINE single key<br>alias registered<br>in dispatcher     |
| <b>PL</b>      | <b>*PLINE</b>               | Draw     | src/modules/draw/draw/pline.rs   | Polyline alias<br>registered<br>directly                 |
| <b>C</b>       | <b>*CIRCLE</b>              | Draw     | src/modules/draw/draw/circle.rs  | Circle single ke<br>alias registered<br>in dispatcher    |
| <b>A</b>       | <b>*ARC</b>                 | Draw     | src/modules/draw/draw/arc.rs     | Arc single key<br>alias registered<br>in dispatcher      |
| <b>REC</b>     | <b>*RECTANG</b>             | Draw     | src/modules/draw/draw/shapes.rs  | Rectangle 3-<br>letter alias<br>registered<br>directly   |
| <b>POL</b>     | <b>*POLYGON</b>             | Draw     | src/modules/draw/draw/shapes.rs  | Polygon alias<br>registered<br>directly                  |
| <b>EL</b>      | <b>*ELLIPSE</b>             | Draw     | src/modules/draw/draw/ellipse.rs | Ellipse alias<br>registered<br>directly                  |
| <b>H</b>       | <b>*HATCH</b>               | Draw     | src/modules/draw/draw/hatch.rs   | Hatch single<br>key alias<br>registered in<br>dispatcher |
| <b>SPL</b>     | <b>*SPLINE</b>              | Draw     | src/modules/draw/draw/spline.rs  | Spline alias<br>registered<br>directly                   |
| <b>XL</b>      | <b>*XLINE</b>               | Draw     | src/modules/draw/draw/xline.rs   | Construction<br>line alias<br>registered<br>directly     |
| <b>RAY</b>     | <b>*RAY</b>                 | Draw     | src/modules/draw/draw/xline.rs   | Ray entity<br>registered                                 |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY | REGISTERED SOURCE MODULE           | REGISTRATION<br>CONTEXT &<br>PURPOSE                          |
|----------------|-----------------------------|----------|------------------------------------|---------------------------------------------------------------|
|                |                             |          |                                    | directly                                                      |
| P0             | *POINT                      | Draw     | src/modules/draw/draw/point.rs     | Point entity<br>alias registered<br>directly                  |
| M              | *MOVE                       | Modify   | src/modules/draw/modify/move.rs    | Move single key<br>alias registered<br>in dispatcher          |
| CO             | *COPY                       | Modify   | src/modules/draw/modify/copy.rs    | Copy primary<br>alias registered<br>directly                  |
| CP             | *COPY                       | Modify   | src/modules/draw/modify/copy.rs    | Secondary<br>legacy copy<br>alias registered<br>independently |
| RO             | *ROTATE                     | Modify   | src/modules/draw/modify/rotate.rs  | Rotate alias<br>registered<br>directly                        |
| SC             | *SCALE                      | Modify   | src/modules/draw/modify/scale.rs   | Scale alias<br>registered<br>directly                         |
| S              | *STRETCH                    | Modify   | src/modules/draw/modify/stretch.rs | Stretch single<br>key alias<br>registered<br>directly         |
| TR             | *TRIM                       | Modify   | src/modules/draw/modify/trim.rs    | Trim alias<br>registered<br>directly                          |
| EX             | *EXTEND                     | Modify   | src/modules/draw/modify/extend.rs  | Extend alias<br>registered<br>directly                        |
| F              | *FILLET                     | Modify   | src/modules/draw/modify/fillet.rs  | Fillet single key<br>alias registered<br>directly             |
| CHA            | *CHAMFER                    | Modify   | src/modules/draw/modify/chamfer.rs | Chamfer alias<br>registered<br>directly                       |
| O              | *OFFSET                     | Modify   | src/modules/draw/modify/offset.rs  | Offset single<br>key alias<br>registered<br>directly          |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY   | REGISTERED SOURCE MODULE           | REGISTRATION<br>CONTEXT &<br>PURPOSE                  |
|----------------|-----------------------------|------------|------------------------------------|-------------------------------------------------------|
| MI             | *MIRROR                     | Modify     | src/modules/draw/modify/mirror.rs  | Mirror alias<br>registered<br>directly                |
| AR             | *ARRAY                      | Modify     | src/modules/draw/modify/array.rs   | Array alias<br>registered<br>directly                 |
| X              | *EXPLODE                    | Modify     | src/modules/draw/modify/explode.rs | Explode single<br>key alias<br>registered<br>directly |
| J              | *JOIN                       | Modify     | src/modules/draw/modify/join.rs    | Join single key<br>alias registered<br>directly       |
| BR             | *BREAK                      | Modify     | src/modules/draw/modify/break.rs   | Break alias<br>registered<br>directly                 |
| E              | *ERASE                      | Modify     | src/modules/draw/modify/erase.rs   | Erase single key<br>alias registered<br>directly      |
| ED             | *TEXTEDIT                   | Annotation | src/app/commands/mod.rs            | Text edit alias<br>registered<br>directly             |
| MT             | *MTEXT                      | Annotation | src/modules/annotate/mtext.rs      | Multiline text<br>alias registered<br>directly        |
| T              | *MTEXT                      | Annotation | src/modules/annotate/mtext.rs      | Single key text<br>alias registered<br>directly       |
| DT             | *TEXT                       | Annotation | src/modules/annotate/text.rs       | Dynamic text<br>alias registered<br>directly          |
| LE             | *QLEADER                    | Annotation | src/modules/annotate/leader.rs     | Quick leader<br>alias registered<br>directly          |
| MLD            | *MLEADER                    | Annotation | src/modules/annotate/mleader.rs    | Multileader alia<br>registered<br>directly            |
| DLI            | *DIMLINEAR                  | Dimension  | src/modules/annotate/dimension.rs  | Linear<br>dimension alias                             |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY               | REGISTERED SOURCE MODULE              | REGISTRATION<br>CONTEXT &<br>PURPOSE                  |
|----------------|-----------------------------|------------------------|---------------------------------------|-------------------------------------------------------|
|                |                             |                        |                                       | registered<br>directly                                |
| <b>DAL</b>     | <b>*DIMALIGNED</b>          | Dimension              | src/modules/annotate/dimension.rs     | Aligned<br>dimension alias<br>registered<br>directly  |
| <b>DRA</b>     | <b>*DIMRADIUS</b>           | Dimension              | src/modules/annotate/dimension.rs     | Radius<br>dimension alias<br>registered<br>directly   |
| <b>DDI</b>     | <b>*DIMDIAMETER</b>         | Dimension              | src/modules/annotate/dimension.rs     | Diameter<br>dimension alias<br>registered<br>directly |
| <b>DAN</b>     | <b>*DIMANGULAR</b>          | Dimension              | src/modules/annotate/dimension.rs     | Angular<br>dimension alias<br>registered<br>directly  |
| <b>DCO</b>     | <b>*DIMCONTINUE</b>         | Dimension              | src/modules/annotate/dimension.rs     | Continue<br>dimension alias<br>registered<br>directly |
| <b>DBA</b>     | <b>*DIMBASELINE</b>         | Dimension              | src/modules/annotate/dimension.rs     | Baseline<br>dimension alias<br>registered<br>directly |
| <b>B</b>       | <b>*BLOCK</b>               | Blocks &<br>References | src/modules/draw/modify/block.rs      | Block single key<br>alias registered<br>directly      |
| <b>I</b>       | <b>*INSERT</b>              | Blocks &<br>References | src/modules/draw/modify/insert.rs     | Insert single key<br>alias registered<br>directly     |
| <b>BE</b>      | <b>*BEDIT</b>               | Blocks &<br>References | src/modules/draw/modify/block_edit.rs | Block edit alias<br>registered<br>directly            |
| <b>PU</b>      | <b>*PURGE</b>               | System &<br>Utilities  | src/app/commands/mod.rs               | Purge alias<br>registered<br>directly                 |
| <b>REN</b>     | <b>*RENAME</b>              | System &<br>Utilities  | src/app/commands/mod.rs               | Rename alias<br>registered<br>directly                |
|                |                             |                        |                                       |                                                       |

| ALIAS<br>TOKEN | CANONICAL<br>OCS<br>COMMAND | CATEGORY             | REGISTERED SOURCE MODULE         | REGISTRATION<br>CONTEXT &<br>PURPOSE                   |
|----------------|-----------------------------|----------------------|----------------------------------|--------------------------------------------------------|
| <b>DI</b>      | <b>*DIST</b>                | Inquiry &<br>Measure | src/modules/draw/inquiry/dist.rs | Distance single<br>alias registered<br>directly        |
| <b>ID</b>      | <b>*ID</b>                  | Inquiry &<br>Measure | src/modules/draw/inquiry/id.rs   | ID point<br>coordinate alias<br>registered<br>directly |



### Application & Clipboard Meta-Commands 39 Entries

Windowing, OS clipboard pipeline (COPYCLIP, PASTE), batch file operations (SAVEALL, CLOSEALL), and terminal console actions.

| COMMAND<br>TOKEN  | ACTION<br>TYPE          | DESCRIPTION &<br>WORKFLOW                                                                        | OCS ENGINE<br>BEHAVIOR                                                          | OCS<br>EQUIVALENT                                                               | COPY                                                                                  |
|-------------------|-------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>COPYCLIP</b>   | Clipboard<br>Primitives | Copies selected drawing objects into the operating system clipboard buffer (DWG vector + image). | Copies internal Entity tree to system pasteboard for cross-window CAD transfer. | Copies internal Entity tree to system pasteboard for cross-window CAD transfer. |  |
| <b>CUTCLIP</b>    | Clipboard<br>Primitives | Copies selected entities to clipboard and immediately erases them from drawing.                  | Atomic copy-then-erase transaction with undo stack record.                      | Atomic copy-then-erase transaction with undo stack record.                      |  |
| <b>PASTE</b>      | Clipboard<br>Primitives | Pastes clipboard geometry into drawing at active cursor coordinate.                              | Direct pasteboard ingestion with interactive dynamic insertion cursor.          | Direct pasteboard ingestion with interactive dynamic insertion cursor.          |  |
| <b>PASTECLIP</b>  | Clipboard<br>Primitives | Pastes clipboard contents into the active space (Model or Layout).                               | Ctrl+V shortcut handler; instantiates pasted entities.                          | Ctrl+V shortcut handler; instantiates pasted entities.                          |  |
| <b>PASTEBLOCK</b> | Clipboard<br>Primitives | Pastes copied entities grouped                                                                   | Wraps entities in internal BlockDefinition                                      | Wraps entities in internal BlockDefinition                                      |  |

| COMMAND<br>TOKEN     | ACTION<br>TYPE          | DESCRIPTION &<br>WORKFLOW                                                                                | OCS ENGINE<br>BEHAVIOR                                                                       | OCS<br>EQUIVALENT                                                                          | COPY                                                                                  |
|----------------------|-------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                      |                         | as an anonymous<br>block reference<br>(*A###).                                                           | before<br>insertion.                                                                         | before<br>insertion.                                                                       |                                                                                       |
| <b>PASTEORIG</b>     | Clipboard<br>Primitives | Pastes clipboard<br>geometry at the<br>exact original XYZ<br>coordinates from<br>which it was<br>copied. | Retains world<br>coordinate<br>transformation<br>matrix without<br>prompting for<br>point.   | Retains world<br>coordinate<br>transformation<br>matrix without<br>prompting for<br>point. |    |
| <b>SAVEALL</b>       | File /<br>Session       | Saves all currently<br>open documents<br>simultaneously.                                                 | Iterates<br>through all<br>open tab<br>buffers and<br>writes changes<br>to disk.             | Iterates through<br>all open tab<br>buffers and<br>writes changes<br>to disk.              |    |
| <b>CLOSEALL</b>      | File /<br>Session       | Closes all open<br>document tabs,<br>prompting to save<br>dirty buffers.                                 | Sequentially<br>invokes close<br>request on all<br>active tab<br>views.                      | Sequentially<br>invokes close<br>request on all<br>active tab<br>views.                    |    |
| <b>CLOSEALLOTHER</b> | File /<br>Session       | Closes all tabs<br>except the<br>currently focused<br>one.                                               | Closes inactive<br>document<br>contexts in<br>application<br>state.                          | Closes inactive<br>document<br>contexts in<br>application<br>state.                        |  |
| <b>NEW</b>           | File /<br>Session       | Creates a new<br>drawing document<br>from default<br>template.                                           | Initializes<br>blank<br>DrawingContext<br>with default<br>layer 0 and<br>standard<br>styles. | Initializes blank<br>DrawingContext<br>with default<br>layer 0 and<br>standard styles.     |  |
| <b>OPEN</b>          | File /<br>Session       | Opens an existing<br>DXF/DWG file from<br>storage.                                                       | Spawns file<br>dialog to parse<br>DXF into<br>memory.                                        | Spawns file<br>dialog to parse<br>DXF into<br>memory.                                      |  |
| <b>CLOSE</b>         | File /<br>Session       | Closes the current<br>drawing document<br>tab.                                                           | Prompts for<br>unsaved edits<br>and unmounts<br>tab.                                         | Prompts for<br>unsaved edits<br>and unmounts<br>tab.                                       |  |
| <b>SAVE</b>          | File /<br>Session       | Saves the current<br>drawing to disk.                                                                    | Serializes<br>active drawing<br>to DXF file.                                                 | Serializes active<br>drawing to DXF<br>file.                                               |  |
| <b>SAVEAS</b>        | File /<br>Session       | Saves current<br>drawing with a                                                                          | Opens save<br>dialog and<br>redirects                                                        | Opens save<br>dialog and<br>redirects                                                      |  |

| COMMAND<br>TOKEN   | ACTION<br>TYPE               | DESCRIPTION &<br>WORKFLOW                                                                        | OCS ENGINE<br>BEHAVIOR                                                  | OCS<br>EQUIVALENT                                                       | COPY                                                                                  |
|--------------------|------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                    |                              | new file name or location.                                                                       | current buffer path.                                                    | current buffer path.                                                    |                                                                                       |
| <b>QSAVE</b>       | File /<br>Session            | Quick saves current drawing without prompting for file path.                                     | Direct file overwrite of active path.                                   | Direct file overwrite of active path.                                   |    |
| <b>EXIT</b>        | File /<br>Session            | Exits the application window after confirming unsaved changes.                                   | Triggers clean shutdown of GUI event loop.                              | Triggers clean shutdown of GUI event loop.                              |    |
| <b>QUIT</b>        | File /<br>Session            | Synonym for EXIT.                                                                                | Graceful application termination.                                       | Graceful application termination.                                       |    |
| <b>SELECTALL</b>   | Selection<br>Meta-<br>Action | Selects all unlocked visible objects in the current drawing space.                               | Ctrl+A handler; pushes all non-frozen entity IDs into selection buffer. | Ctrl+A handler; pushes all non-frozen entity IDs into selection buffer. |    |
| <b>DESELECT</b>    | Selection<br>Meta-<br>Action | Clears the current object selection set.                                                         | Empties active selection hashset.                                       | Empties active selection hashset.                                       |  |
| <b>DESELALL</b>    | Selection<br>Meta-<br>Action | Synonym for clearing selection.                                                                  | Deselects all entities.                                                 | Deselects all entities.                                                 |  |
| <b>ADDSELECTED</b> | Selection<br>Meta-<br>Action | Starts the creation command corresponding to selected object with matching layer and properties. | Inspects selected entity type and starts matching tool.                 | Inspects selected entity type and starts matching tool.                 |  |
| <b>CLEAR</b>       | Terminal &<br>Shell          | Clears the scrollbar history buffer of the command line window.                                  | Resets the command output text buffer.                                  | Resets the command output text buffer.                                  |  |
|                    |                              |                                                                                                  |                                                                         |                                                                         |                                                                                       |

| COMMAND<br>TOKEN | ACTION<br>TYPE        | DESCRIPTION &<br>WORKFLOW                                                        | OCS ENGINE<br>BEHAVIOR                                               | OCS<br>EQUIVALENT                                                    | COPY                                                                                  |
|------------------|-----------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>CLR</b>       | Terminal &<br>Shell   | Shorthand alias<br>for CLEAR.                                                    | Clears terminal<br>text.                                             | Clears terminal<br>text.                                             |    |
| <b>HELP</b>      | Help &<br>Diagnostics | Opens<br>documentation or<br>help viewer.                                        | Launches in-app<br>command<br>reference or<br>documentation<br>tab.  | Launches in-<br>app command<br>reference or<br>documentation<br>tab. |    |
| <b>ABOUT</b>     | Help &<br>Diagnostics | Displays<br>application build<br>version, commit<br>hash, and engine<br>credits. | Renders version<br>and compiler<br>flags modal.                      | Renders version<br>and compiler<br>flags modal.                      |    |
| <b>CHANGELOG</b> | Help &<br>Diagnostics | Displays commit<br>history and recent<br>feature release<br>notes.               | Opens markdown<br>release notes<br>window.                           | Opens<br>markdown<br>release notes<br>window.                        |    |
| <b>PERF</b>      | Help &<br>Diagnostics | Toggles rendering<br>performance HUD<br>(FPS, draw calls,<br>GPU memory).        | Activates live<br>frame-timer<br>overlay.                            | Activates live<br>frame-timer<br>overlay.                            |   |
| <b>SHORTCUTS</b> | Help &<br>Diagnostics | Opens shortcut<br>manager and<br>keyboard<br>keybindings<br>dialog.              | Displays<br>interactive<br>keybindings<br>overlay.                   | Displays<br>interactive<br>keybindings<br>overlay.                   |  |
| <b>CASCADE</b>   | Terminal &<br>Shell   | Arranges open<br>drawing document<br>windows in a<br>cascaded stack.             | Positions<br>internal MDI<br>windows in<br>cascade layout.           | Positions<br>internal MDI<br>windows in<br>cascade layout.           |  |
| <b>TILEHORZ</b>  | Terminal &<br>Shell   | Arranges open<br>drawing windows<br>stacked<br>horizontally.                     | Splits main<br>viewport<br>vertically into<br>horizontal<br>strips.  | Splits main<br>viewport<br>vertically into<br>horizontal<br>strips.  |  |
| <b>TILEVERT</b>  | Terminal &<br>Shell   | Arranges open<br>drawing windows<br>side by side<br>vertically.                  | Splits main<br>viewport<br>horizontally<br>into vertical<br>columns. | Splits main<br>viewport<br>horizontally into<br>vertical<br>columns. |  |
|                  |                       |                                                                                  |                                                                      |                                                                      |                                                                                       |

| COMMAND<br>TOKEN     | ACTION<br>TYPE               | DESCRIPTION &<br>WORKFLOW                                                 | OCS ENGINE<br>BEHAVIOR                                               | OCS<br>EQUIVALENT                                                    | COPY                                                                                  |
|----------------------|------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>ARRANGE</b>       | Terminal &<br>Shell          | Arranges<br>minimized<br>document icons<br>across bottom of<br>workspace. | Aligns<br>minimized tab<br>icons.                                    | Aligns<br>minimized tab<br>icons.                                    |    |
| <b>LAYOUTPANEL</b>   | Terminal &<br>Shell          | Toggles visibility<br>of Paper Space<br>layout<br>management<br>panel.    | Shows/hides<br>layout dock<br>widget.                                | Shows/hides<br>layout dock<br>widget.                                |    |
| <b>LAYOUTTAB</b>     | Terminal &<br>Shell          | Toggles display of<br>bottom layout tab<br>strip.                         | Toggles tab bar<br>widget<br>visibility.                             | Toggles tab bar<br>widget visibility.                                |    |
| <b>PRINTALL</b>      | File /<br>Session            | Triggers batch<br>print sequence for<br>all sheets in<br>drawing.         | Runs print<br>pipeline across<br>all layouts.                        | Runs print<br>pipeline across<br>all layouts.                        |    |
| <b>QUICKPRINT</b>    | File /<br>Session            | Sends active<br>display window<br>directly to default<br>printer driver.  | Rasterizes<br>current canvas<br>viewport and<br>sends to<br>printer. | Rasterizes<br>current canvas<br>viewport and<br>sends to<br>printer. |  |
| <b>PLUGINMANAGER</b> | Help &<br>Diagnostics        | Opens plugin and<br>extension<br>management<br>window.                    | Inspects<br>dynamically<br>loaded Rust<br>plugins.                   | Inspects<br>dynamically<br>loaded Rust<br>plugins.                   |  |
| <b>PLUGINS</b>       | Help &<br>Diagnostics        | Shorthand alias<br>for<br>PLUGINMANAGER.                                  | Opens plugin<br>manager.                                             | Opens plugin<br>manager.                                             |  |
| <b>UNGROUP</b>       | Selection<br>Meta-<br>Action | Disbands an<br>object selection<br>group into<br>independent<br>entities. | Removes group<br>container and<br>releases child<br>entities.        | Removes group<br>container and<br>releases child<br>entities.        |  |



### 3D & Mesh Commands 28 Commands

OCS is a full 2D/3D CAD application, these tokens are part of its native 3D modeling, surfacing, and solid-editing command set.



**Native 3D Support:** OCS's Rust engine implements 3D solid and surface operations directly (dynamic press-pulling, boolean union/subtract/intersect, sweep/revolve, and polygon meshing). A small number of these 28 legacy AutoCAD tokens are not yet implemented — see each row's note below.

| COMMAND<br>TOKEN | CATEGORY           | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                | IMPLEMENTATION<br>NOTE                                                              | COPY |
|------------------|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|
| 3DMESH           | 3D Mesh            | ● Implemented | <p>Builds complex organic terrain or curved 3D mesh shells.</p> <p>Creates a free-form 3D polygon mesh defined by an M x N grid of 3D coordinate vertices.</p>    | Implemented as a native 3D polygon-mesh entity in OCS's own drawing engine.         |      |
| 3DFACE           | 3D Mesh            | ● Implemented | <p>Fundamental legacy 3D surface facet entity.</p> <p>Creates a 3-sided or 4-sided surface planar polygon anywhere in 3D space with optional invisible edges.</p> | Implemented as a native 3D surface facet entity, including invisible-edge support.  |      |
| 3DPOLY           | 3D Surface & Curve | ● Implemented | <p>Draws 3D pipe paths, topographic contour lines, and structural wireframes.</p> <p>Creates a 3D polyline with straight segments whose vertices can have</p>     | Implemented as a native 3D polyline entity with independent per-vertex Z elevation. |      |

| COMMAND<br>TOKEN | CATEGORY                     | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                                   | IMPLEMENTATION<br>NOTE                                                                                             | COPY                                                                                  |
|------------------|------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                  |                              |               | <p>differing Z elevations.</p>                                                                                                                                                       |                                                                                                                    |                                                                                       |
| PRESSPULL        | Solid Modeling / Direct Edit | ● Implemented | <p>Direct modeling tool to punch holes or raise bosses in 3D solids.</p> <p>Dynamically extrudes or offsets 2D enclosed boundaries and planar faces of 3D solids by cursor drag.</p> | Implemented with OCS's own solid-modeling engine – dynamic extrude/offset of enclosed boundaries and planar faces. |    |
| SWEEP            | Solid Modeling / Direct Edit | ● Implemented | <p>Models ducts, pipes, wires, moldings, and railings in 3D.</p> <p>Extrudes a 2D profile curve along an open or closed 2D/3D trajectory path curve.</p>                             | Implemented natively – profile sweep along open or closed 2D/3D trajectory curves.                                 |  |
| REVOLVE          | Solid Modeling / Direct Edit | ● Implemented | <p>Designs turned components: shafts, pulleys, spheres, cones, bottles.</p> <p>Creates a 3D solid or surface by sweeping a 2D planar object around an axis of rotation.</p>          | Implemented natively – solid/surface generation by revolving a profile around an axis.                             |  |
| UNION            | Solid Modeling / Direct Edit | ● Implemented | Standard constructive solid                                                                                                                                                          | Implemented as a native boolean solid union in                                                                     |  |

| COMMAND<br>TOKEN | CATEGORY                     | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                    | IMPLEMENTATION<br>NOTE                                                                           | COPY                                                                                  |
|------------------|------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                  |                              |               | <p>geometry (CSG) boolean union.</p> <p>Combines two or more 3D solids, surfaces, or 2D regions into a single composite object by boolean addition.</p>               | OCS's own geometry kernel.                                                                       |                                                                                       |
| SUBTRACT         | Solid Modeling / Direct Edit | ● Implemented | <p>Cuts slots, holes, and voids out of solid models.</p> <p>Removes the volume of selected 3D solids or regions from another primary solid by boolean difference.</p> | Implemented as a native boolean solid difference operation.                                      |    |
| INTERSECT        | Solid Modeling / Direct Edit | ● Implemented | <p>CSG boolean intersection.</p> <p>Creates a composite 3D solid or region from the overlapping common volume of two or more objects.</p>                             | Implemented as a native boolean solid intersection operation.                                    |  |
| SECTION          | Solid Modeling / Direct Edit | ● Implemented | <p>Generates 2D profile cuts through 3D assemblies.</p> <p>Uses the intersection of a cutting plane and 3D solids to generate a</p>                                   | Implemented natively – generates a 2D cross-sectional Region from a cutting plane and 3D solids. |  |

| COMMAND<br>TOKEN | CATEGORY                           | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                                            | IMPLEMENTATION<br>NOTE                                                                              | COPY                                                                                  |
|------------------|------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                  |                                    |               | 2D cross-<br>sectional<br>Region object.                                                                                                                                                      |                                                                                                     |                                                                                       |
| <b>INTERFERE</b> | Solid<br>Modeling /<br>Direct Edit | ● Implemented | Clash<br>detection<br>between<br>structural and<br>mechanical<br>parts.<br>Highlights 3D<br>solid clashing<br>and generates<br>temporary<br>interference<br>solids at points<br>of collision. | Implemented<br>natively –<br>spatial collision<br>detection between<br>selected 3D<br>solids.       |    |
| <b>SOLID2D</b>   | 3D Mesh                            | ● Implemented | Draws filled<br>planar<br>polygons with<br>triangular or<br>bow-tie<br>vertices.<br>Creates solid-<br>filled polygonal<br>triangular and<br>quadrilateral<br>planar faces.                    | Implemented as a<br>native filled-<br>polygon entity<br>(triangular and<br>quadrilateral<br>faces). |  |
| <b>S0</b>        | 3D Mesh                            | ● Implemented | Draws filled<br>2D planar<br>solids.<br>Shorthand alias<br>for SOLID2D<br>command.                                                                                                            | Registered as a<br>genuine alias of<br>SOLID2D.                                                     |  |
| <b>TRACE</b>     | 3D Mesh                            | ● Implemented | Legacy<br>precursor to<br>wide polylines<br>(PLINE).<br>Legacy drafting<br>command that<br>creates thick<br>solid line<br>segments with<br>mitred<br>endpoints.                               | Implemented as a<br>legacy thick-<br>line-segment<br>entity, kept for<br>script<br>compatibility.   |  |

| COMMAND<br>TOKEN | CATEGORY         | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                  | IMPLEMENTATION<br>NOTE                                                       | COPY                                                                                  |
|------------------|------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 3DMOVE           | 3D<br>Transforms | ● Implemented | <p>Interactive 3D transformation triad with axis locking.</p> <p>Displays a 3D move coordinate gizmo to constrain translation strictly along X, Y, or Z axes.</p>   | Implemented as a native 3D move gizmo with axis-locked translation.          |    |
| 3DROTATE         | 3D<br>Transforms | ● Implemented | <p>Rotates solids around X, Y, or Z gimbal rings.</p> <p>Displays 3D rotation gizmo rings to constrain rotation around a specific coordinate axis in space.</p>     | Implemented as a native 3D rotation gizmo with axis-locked rings.            |    |
| 3DSCALE          | 3D<br>Transforms | ● Planned     | <p>Non-uniform or uniform 3D scaling.</p> <p>Displays a 3D scale gizmo to resize objects along a plane or axis.</p>                                                 | Not yet implemented – the 3D scale gizmo has no dedicated tool yet.          |  |
| 3DARRAY          | 3D<br>Transforms | ● Implemented | <p>Multi-level structural repetition (floors, grids).</p> <p>Creates a 3-dimensional rectangular array (rows, columns, levels) or polar array around a 3D axis.</p> | Implemented natively, including 3D (Z-level) rectangular and polar arraying. |  |

| COMMAND<br>TOKEN | CATEGORY                           | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                         | IMPLEMENTATION<br>NOTE                                                                                  | COPY                                                                                  |
|------------------|------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ARRAY3D          | 3D<br>Transforms                   | ● Implemented | Multi-tier 3D<br>arraying.<br>Synonym for<br>3DARRAY<br>command.                                                                                                           | Registered as a<br>genuine alias of<br>3DARRAY.                                                         |    |
| ROTATE3D         | 3D<br>Transforms                   | ● Implemented | Precision non-<br>orthogonal 3D<br>alignment.<br>Rotates objects<br>in 3D space<br>around any<br>arbitrary two-<br>point line axis.                                        | Implemented<br>natively –<br>arbitrary-axis 3D<br>rotation about a<br>two-point line.                   |    |
| CONVTOSOLID      | Solid<br>Modeling /<br>Direct Edit | ● Planned     | Upgrades<br>mesh/surface<br>geometry into<br>solid objects.<br>Converts<br>eligible 3D<br>surfaces,<br>meshes, and<br>closed<br>watertight<br>polylines into<br>3D solids. | Not yet<br>implemented –<br>watertight<br>mesh/surface-to-<br>solid conversion<br>is planned.           |   |
| CONVTSURFACE     | 3D Surface<br>& Curve              | ● Implemented | Generates<br>NURBS or<br>analytic<br>surface<br>geometry.<br>Converts 2D<br>objects, faces,<br>and meshes<br>into 3D<br>surfaces.                                          | Implemented<br>natively –<br>converts eligible<br>2D objects,<br>faces, and meshes<br>into 3D surfaces. |  |
| CONVTOMESH       | 3D Mesh                            | ● Planned     | Prepares CAD<br>geometry for<br>organic<br>polygonal<br>mesh<br>sculpting.<br>Converts 3D<br>solids and<br>surfaces into                                                   | Not yet<br>implemented –<br>solid/surface-to-<br>subdivision-mesh<br>conversion is<br>planned.          |  |

| COMMAND<br>TOKEN | CATEGORY              | OCS STATUS | OCS 3D<br>FUNCTION                                                                                                                                                                                          | IMPLEMENTATION<br>NOTE                                                                                                        | COPY                                                                                  |
|------------------|-----------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                  |                       |            | <p>editable<br/>subdivision<br/>polygon<br/>meshes.</p>                                                                                                                                                     |                                                                                                                               |                                                                                       |
| EDGESURF         | 3D Surface<br>& Curve | ● Planned  | <p>Creates<br/>bicubic<br/>surface mesh<br/>between four<br/>boundary<br/>curves.</p> <p>Constructs a<br/>3D Coons<br/>surface patch<br/>from four<br/>contiguous<br/>touching<br/>boundary<br/>curves.</p> | <p>Not yet<br/>implemented –<br/>Coons-patch<br/>surface<br/>generation from<br/>four boundary<br/>curves is<br/>planned.</p> |    |
| REVSURF          | 3D Surface<br>& Curve | ● Planned  | <p>Rotational<br/>surface mesh<br/>creation.</p> <p>Creates a 3D<br/>faceted mesh<br/>surface of<br/>revolution by<br/>rotating a<br/>profile curve<br/>around an axis.</p>                                 | <p>Not yet<br/>implemented –<br/>mesh surface of<br/>revolution is<br/>planned.</p>                                           |  |
| RULESURF         | 3D Surface<br>& Curve | ● Planned  | <p>Ruled mesh<br/>connecting<br/>two parallel or<br/>skewed profile<br/>curves.</p> <p>Creates a 3D<br/>ruled surface<br/>mesh between<br/>two defining<br/>curves.</p>                                     | <p>Not yet<br/>implemented –<br/>ruled surface<br/>interpolation<br/>between two<br/>curves is<br/>planned.</p>               |  |
| TABSURF          | 3D Surface<br>& Curve | ● Planned  | <p>Extruded open<br/>mesh<br/>surfaces.</p> <p>Creates a<br/>tabulated<br/>cylinder 3D<br/>mesh from a</p>                                                                                                  | <p>Not yet<br/>implemented –<br/>tabulated<br/>cylinder surface<br/>generation is<br/>planned.</p>                            |  |

| COMMAND<br>TOKEN | CATEGORY                           | OCS STATUS    | OCS 3D<br>FUNCTION                                                                                                                                                                                                          | IMPLEMENTATION<br>NOTE                                                                               | COPY                                                                                |
|------------------|------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                  |                                    |               | path curve and<br>a direction<br>vector.                                                                                                                                                                                    |                                                                                                      |                                                                                     |
| FLATSHOT         | Solid<br>Modeling /<br>Direct Edit | ● Implemented | Generates 2D<br>architectural<br>elevations and<br>plans directly<br>from 3D<br>models.<br>Projects all 3D<br>solid geometry<br>into a 2D<br>flattened<br>vector drawing<br>based on the<br>active viewport<br>perspective. | Implemented<br>natively –<br>projects 3D solid<br>geometry into a<br>flattened 2D<br>vector drawing. |  |